

- **Save additionally:** The project is **additionally** saved on the storage medium (as *.zw1) or, depending on the storage medium selected, is set up as an e-mail attachment. The project to be backed up still remains in the original project path and is neither deleted nor changed. This is the option that I recommend.
- **File off for external editing:** The project is also additionally saved (as *.zw1) and the original project is write-protected so that no further changes are possible. This method is useful when the project (e.g.) is to be filed off to a customer because he/she wishes to make changes and no other changes should be made to the original project until it is restored.
- **Archive:** The project is not additionally backed up; it is filed off to a different drive (as *.zw1). The original project, except for the project information (*ProjectInfo.xml*), is deleted from the original project path.

Backup method

In addition to the selection of the backup method, EPLAN also provides two backup media options for the project to be backed up.

Backup medium

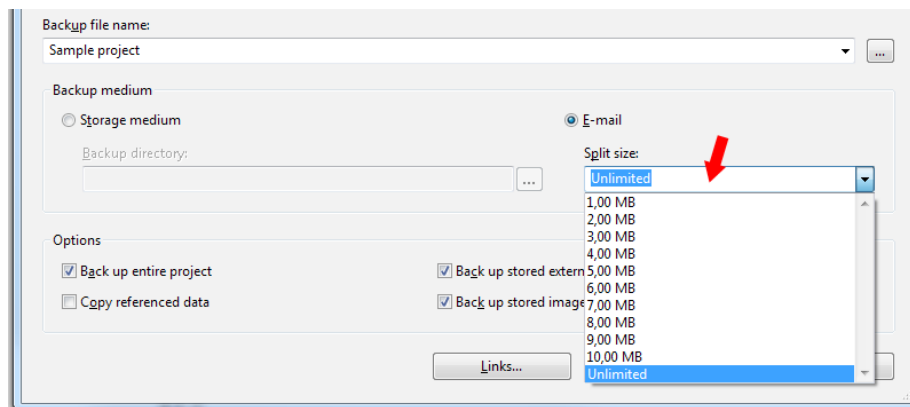


Fig. 9.9 Backup medium selection

- **Storage medium:** Usually another drive – you can use any directory structure with absolutely no limitations. If supported by the operating system, the project can be directly backed up to a CD/DVD burner.
- **E-mail:** The project is backed up and given a *.zw1 file extension. The installed e-mail client is then opened and the project added as an attachment. EPLAN provides the option of dividing the e-mail file into several files (parts) (some variables have been preset via a selection list, but these can be changed in any way; that is, even 6.5 MB is possible). This is useful when the recipient cannot receive large attachments. The recipient **must** then save all the files (parts) in the received e-mails in the same directory in order to restore the project later on.

A project was backed up using the **Save additionally** method and the **E-mail** backup medium. The split size was set to 2 MB.



Backup via e-mail

When you click **OK**, EPLAN starts the data backup. EPLAN then generates several files internally, opens the installed e-mail program and attaches the first part (Part 3 of [Total]) to a new e-mail.

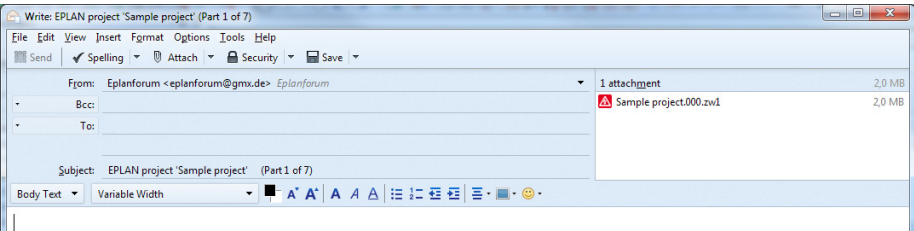


Fig. 9.10 First e-mail generated automatically by EPLAN

The following information is automatically entered as the subject: the project name, information on which part is attached, and the total number of parts to be sent.

EPLAN does not continue with the data backup until the e-mail has been sent. In this way, EPLAN automatically generates all parts and the corresponding e-mails. You only need to enter the recipient and send the e-mail.

The last part (Part 1 of [Total]) is the most important part. When restoring the project later on, EPLAN first looks for this part in the selected directory, and all the other parts must also be in this directory.

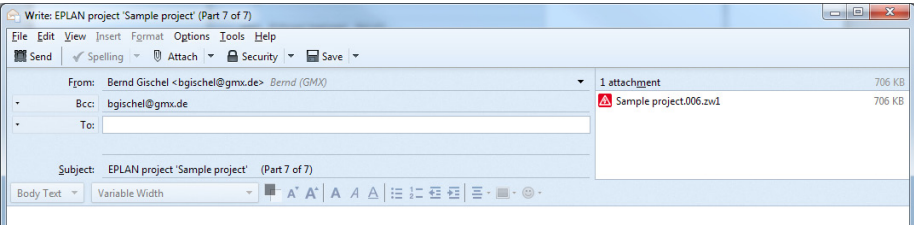


Fig. 9.11 “Last” e-mail containing the last part of the project backup

This completes the e-mail backup. After the final part has been sent, EPLAN displays a success message.

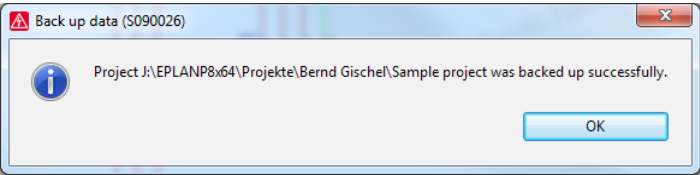


Fig. 9.12 Message that all parts have been successfully sent



NOTE: If errors exist in any of these parts, then EPLAN cannot restore the project later on. If this occurs, then EPLAN displays a message window telling the user that the project cannot be restored. In this case, the only solution is to request another data backup.

EPLAN writes the reason for cancelling in the system messages. The reason can be viewed by clicking the **SYSTEM MESSAGES...** link. EPLAN opens the **System messages** dialog. System messages can be saved or you can close the dialog. If you close the **System messages** dialog, you are returned to the previous dialog. Simply click OK to return to graphical editing.

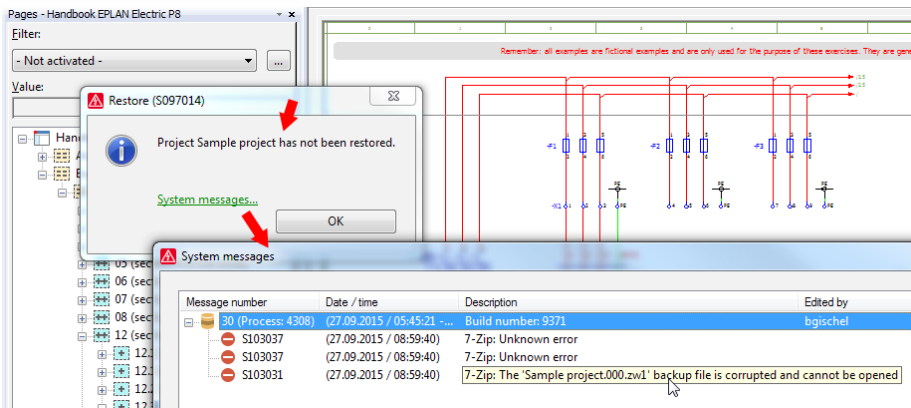


Fig. 9.13 Error message when the project could not be restored

To prevent a backed-up project from becoming unnecessarily “big” (data volume), EPLAN offers the user four additional options in the **Back up projects** dialog to reduce the data volume.

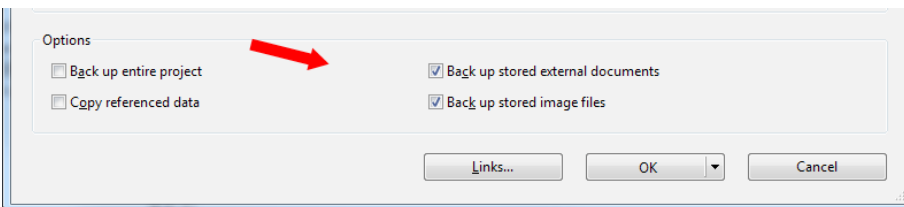


Fig. 9.14 Reduce data volume using these options

- **Back up entire project:** Backs up the entire project, including all data in the project directory and all subdirectories.
- **Copy referenced data:** If this option is set, EPLAN backs up everything, including referenced data. This increases the amount of data in the project. The options for stored external documents and stored image files must also be activated.



NOTE: Referenced data is data that is not stored in the project, but is available only as reference. In short: EPLAN only knows the link (file path) to where, for example, the graphic is stored on the server.

- **Back up stored external documents:** This option allows the user to include in or exclude from the data backup external documents stored in the project, such as PDF or Excel documents. However, only the external documents in the `\[project name]\DOC` directory are included in the backup.
- **Back up stored image files:** Similar to the previous option (external documents), you can define whether image files are included in or excluded from the backup process. This also only applies to image files in the `\[project name]\Images` directory.

External documents or image files that are **not** in the project directory will **not** be automatically included in a backup. The previously mentioned **Back up external documents** and **Back up image files** options have **no effect** at this point.

If these files should also be backed up, then they must either be stored in the project (copied into the project directory on insertion during project editing) or they must be additionally and automatically stored in the project directory by EPLAN via the **LINKS** button.

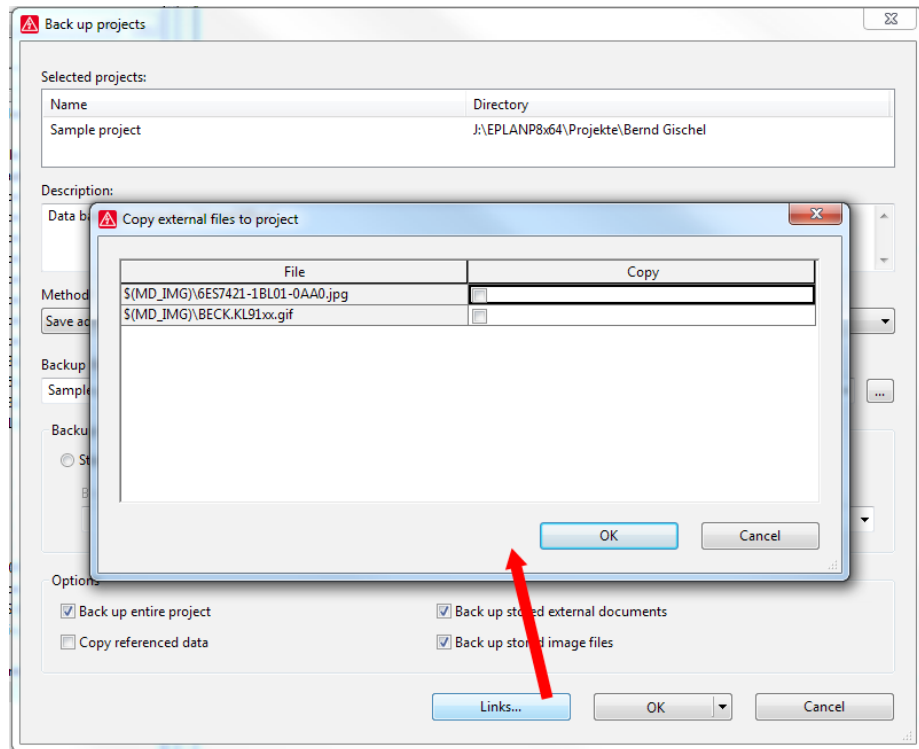


Fig. 9.15 Copying externally linked files to the project (back up projects)

For external documents and image files that are linked into the project via hyperlinks, there is another way of collecting them and copying them together into the project directory before performing a data backup. You use the **EDIT/OTHER/LINKS** menu to open the **Copy external files to project** dialog. This dialog is identical to the one that you can call up during the backup of projects.

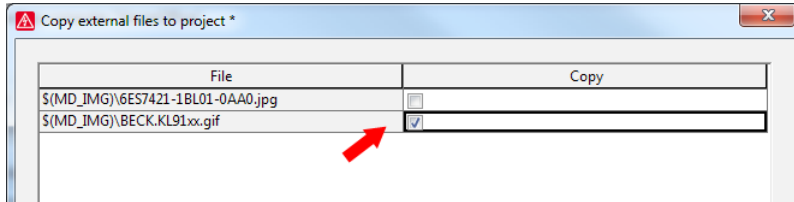


Fig. 9.16 Copying externally linked files to the project (graphical editor)

EPLAN now lists all linked external documents in this dialog. You then select the **Copy** function for all desired documents. When you confirm the dialog by clicking **OK**, all selected files are copied to the `\[project name]\DOC` or `\IMAGES` project directory.

9.2.2 Restore projects

Backed-up projects can be restored. First call the **PROJECT/RESTORE/PROJECT** menu and then select in graphical editing one or several projects you wish to restore (restore back).

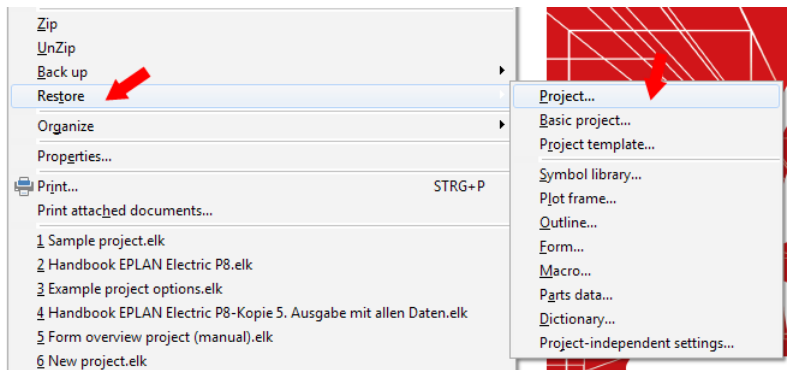


Fig. 9.17 Restore projects menu

The **Restore project** dialog opens with a number of setting and selection options.

Restore projects
settings

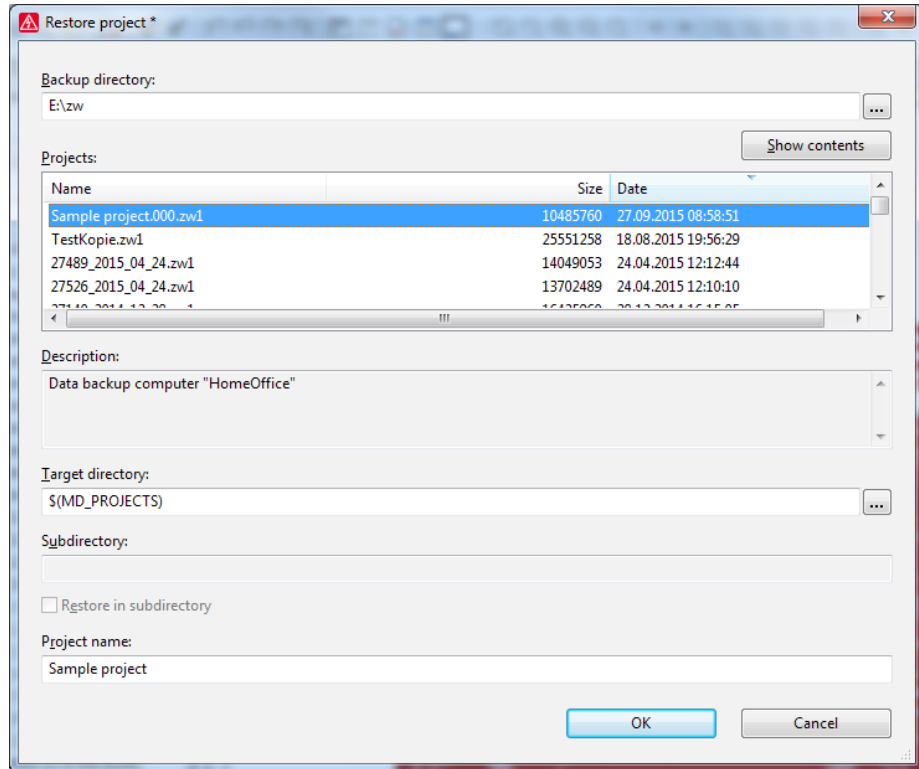


Fig. 9.18 Restore project dialog

- **Backup directory:** In the **Restore project** dialog, you can select the backup directory via the [...] button. You can use any type of backup directory, which is not bound to any specific structure.



TIP: It is a good idea to define a permanent directory for the backup files and/or the e-mail files. It is also a good idea to integrate this directory into the normal EPLAN directory used for the installation.

- **Projects:** Here, you need to select the project to be restored.
- **Target directory:** Before restoring a project, you should first check the *target directory* field and, if necessary, set it to the correct project path. To change the target directory, click the [...] button to select a different directory.
- **Project name:** In this field, you can assign a new project name before you restore the project, or you can leave the current project name as is.

Click OK or a double-click the project (in the *Projects* field) to restore the project using the set parameters. EPLAN then begins the restore process. No further query dialogs are displayed.

If a project of the same name already exists in the target directory, then EPLAN displays a warning message asking if the target project should be overwritten. If you click the YES button, EPLAN restores the project and overwrites the old project.

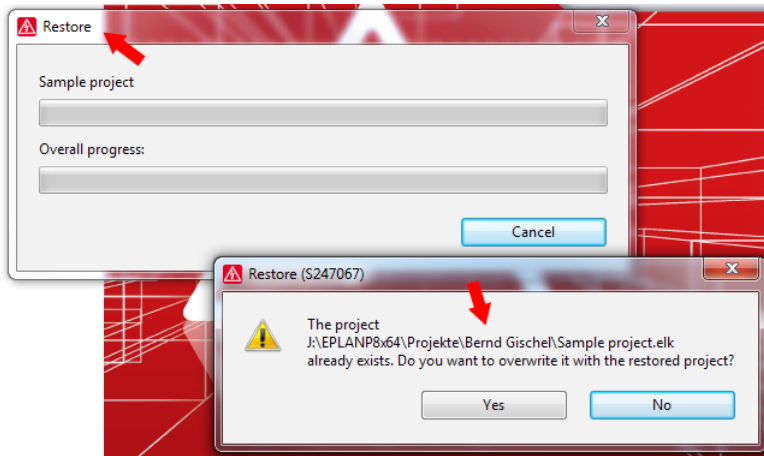


Fig. 9.19

Possible message if a project already exists

EPLAN can also overwrite opened projects if you wish. After a prompt, EPLAN closes the project and then restores it.

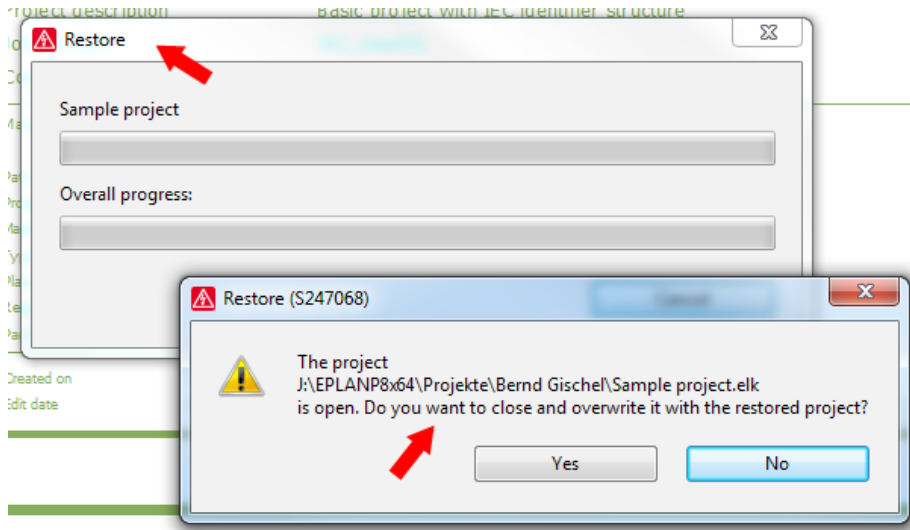


Fig. 9.20

Another message if a project already exists

Once the project has been fully restored, EPLAN opens it in the **page navigator**.

If you click the NO button, EPLAN cancels the restore process and displays a message stating that the project was not restored.

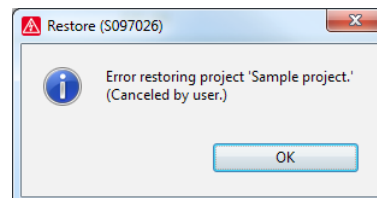


Fig. 9.21

Note displayed when canceled by user

■ 9.3 Other important settings

There are also default settings related to the backing up and restoring of data, such as for defining backup directories or for cleaning projects of old data that is no longer required. It is also possible to automate the process of backing up data and cleaning projects.

9.3.1 Default settings for project backup (global user setting)

You can define global default settings for projects. These are then applied to the specific project when it is backed up for the first time. You can adjust these and other default settings in EPLAN via **OPTIONS/SETTINGS/USER/DATA BACKUP/DEFAULT SETTINGS/PROJECTS**. Here you can set the target directory, the split size of e-mail attachments and other options such as back up entire project, save external documents and image files, and the default setting for the backup description field. These settings are then applied when a project is backed up.

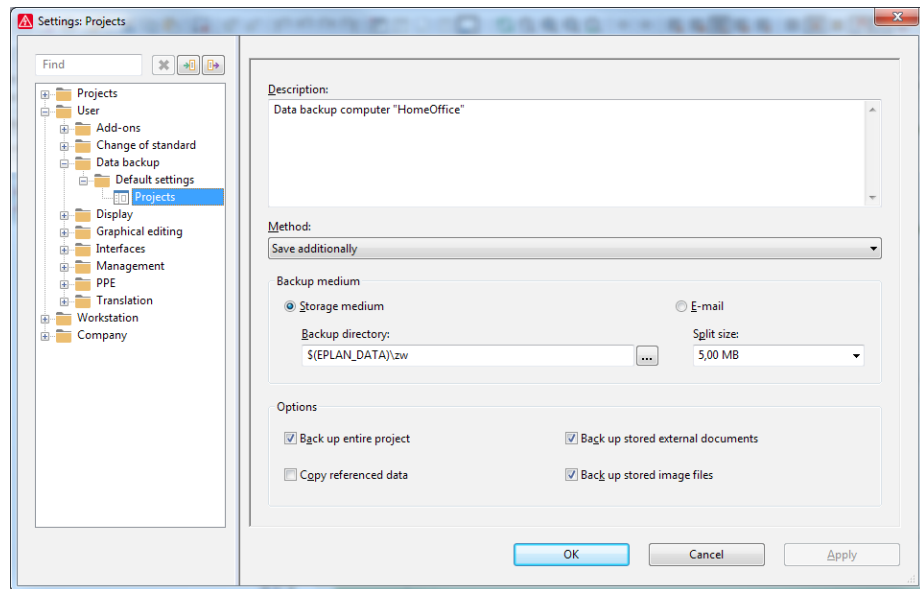


Fig. 9.22
Default settings for
backing up projects



TIP: If you subsequently change these settings yourself when backing up a project, EPLAN remembers these settings precisely for this project. The global user settings then no longer apply. The updated data backup settings for a project always have priority over the data backup settings in the user settings and will always be used in the future for this project.

The backup directory and the split size of e-mail attachments can, of course, be changed later, also for projects that have already been backed up. The new settings are then applied again for the next backup process.

9.3.2 Compress project (remove unnecessary data)

While editing a project, it can become “filled” with a great deal of data. Unplaced devices may be inserted or forms (master data) may be created for test purposes only.

Since EPLAN stores all data in the project automatically at first, even when used only once, a project may end up containing, for example, ten different terminal diagram forms or dozens of parts even though only one of these was actually used for reports at the end of project editing or the parts are no longer needed.

Data that has become unnecessary in the course of project editing can be removed in EPLAN via the *Compress project* function. You access this function via the **PROJECT/ORGANIZE/COMPRESS MENU**.

Compress project function

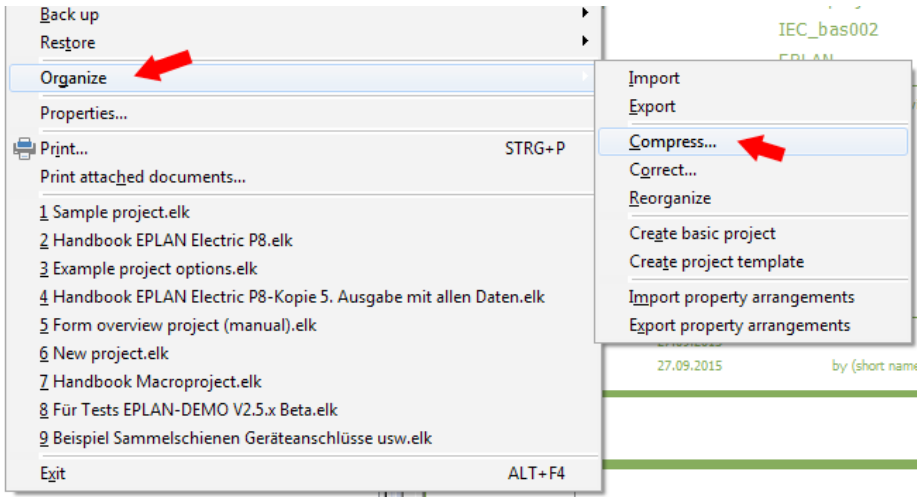


Fig. 9.23 Project/Organize/Compress menu

EPLAN then starts the **Compress project** dialog.

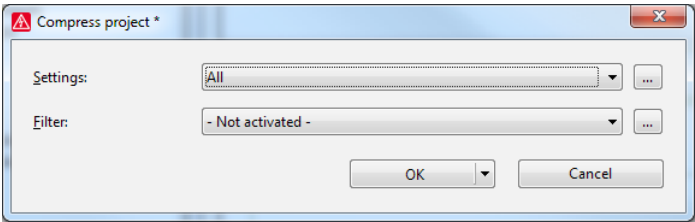


Fig. 9.24 Compress project dialog

The dialog offers several ways of affecting the data volume of the project. You select the desired compression scheme to compress the data in the **Settings** selection field. EPLAN provides a number of default settings (finished schemes, which can be selected from a selection list), but here too, you can also define your own personal settings by creating and saving your own schemes. Click the [...] button next to the **Settings** selection field. The **Settings: Compression** dialog opens.

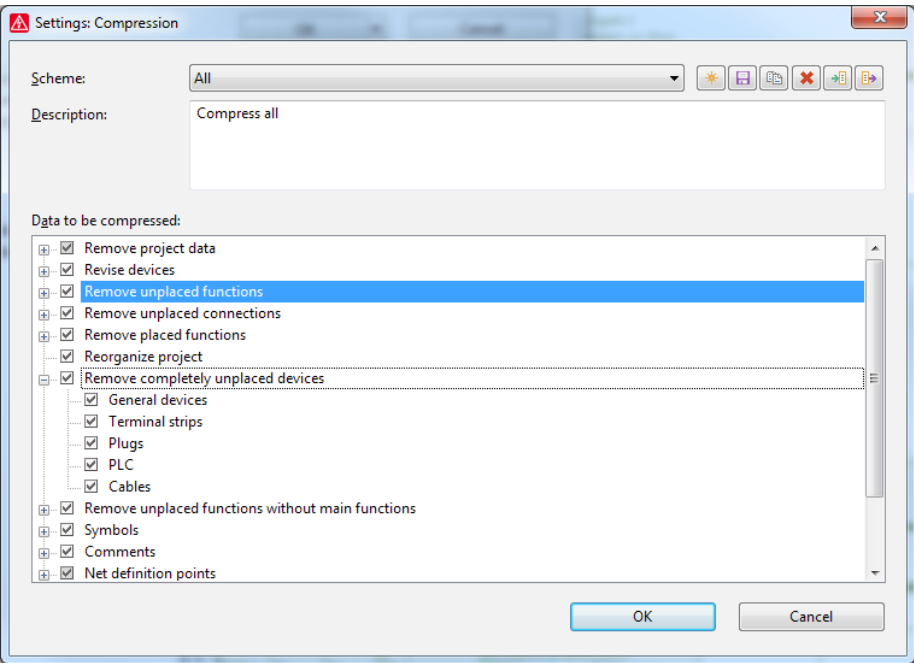


Fig. 9.25
Settings options for a
compression scheme

Here, you can use the familiar graphical buttons to create a new scheme or copy and change an existing scheme.

In the lower **Data to be compressed** field, EPLAN offers a number of actions that can be performed during the compression.

One example is the item **Remove project data/Unused forms**. This setting removes all unused forms from the project and the project then contains only forms that are actually used.

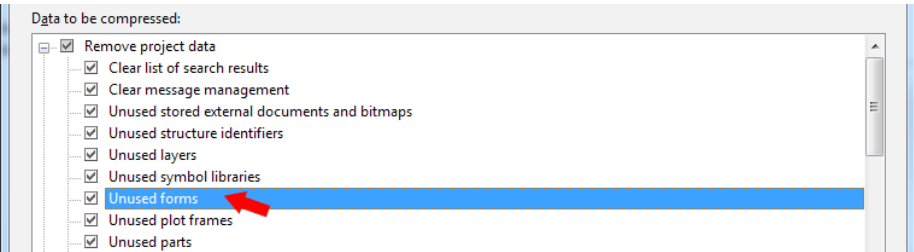


Fig. 9.26
Setting to remove
unnecessary forms



NOTE: If, for example, forms documentation was created before the compression, all forms (including the unused ones) remain stored in the project because they are needed for the forms documentation. You should remove the reports here, especially the forms documentation, prior to compression.

The other, second, method is to use filters for the compression process. As usual, filters can be set and are either created or edited as schemes. In the **Filter** for compression you can, for example, exclude specific areas from the compression process, such as particular functions in the schematic or entire structure identifiers.

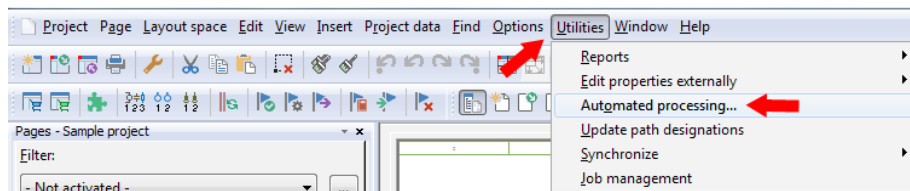
EPLAN thus provides various options for specifying what data is to be removed (using filters) and what areas or data the compression process should include.

9.3.3 Automated processing of a project

So far, we have looked only at individual steps. The **Organize/Compress** function was used to free the project of unnecessary ballast and the project was then backed up using the **Project/Back up/Projects** function.

The **Automated processing** function in EPLAN allows such actions to be combined together and automated. In this example, we will use the compress project and data backup actions.

The **Automated processing** function is accessed via the **UTILITIES/AUTOMATED PROCESSING** menu.



Automated processing with scripts

Fig. 9.27 The Automated processing menu item

Afterwards, EPLAN opens the **Run: Automated processing** dialog.

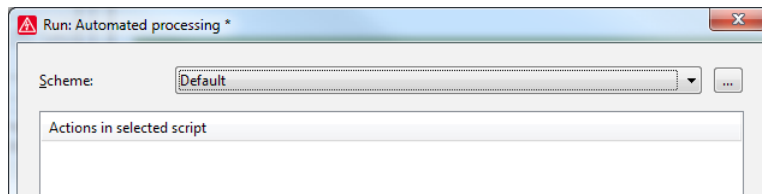


Fig. 9.28 Run: Automated processing dialog

Here, only an existing scheme can be selected, or a new one created.

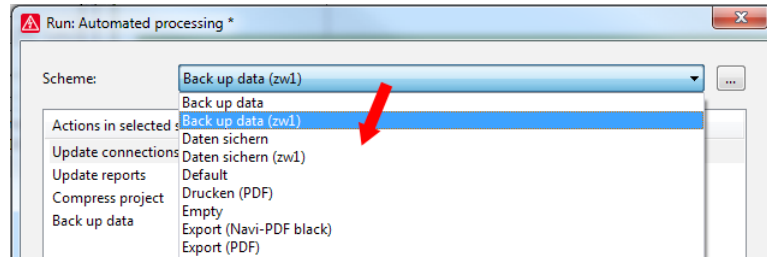


Fig. 9.29 Scheme drop-down list

The [...] button contains the **Settings: Automated processing** dialog. As is customary in EPLAN, you can create and save new schemes in this dialog. All projects can use these schemes.

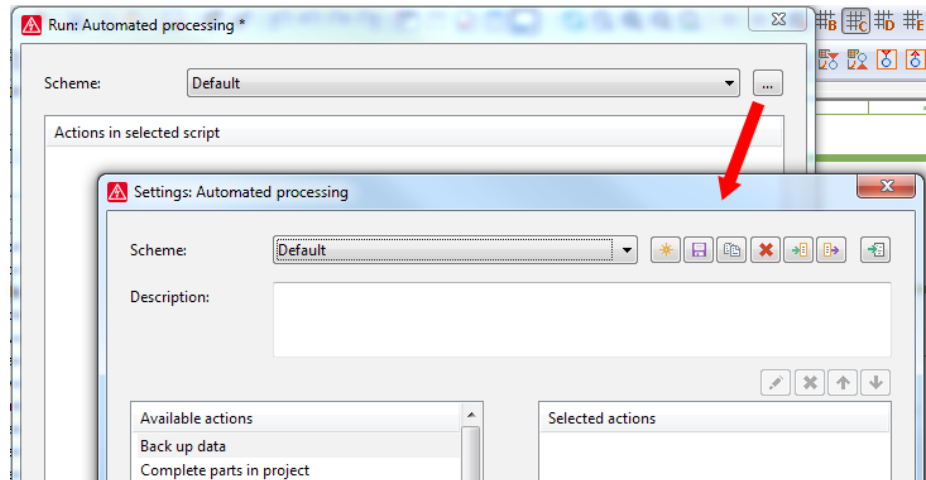


Fig. 9.30 The possible settings of automated processing

The **Automated processing** function is a scheme, as already mentioned. The Automated processing function can be used to export and import schemes. The schemes are in the *Scripts* directory (applies for a standard installation). The dialog contains the **Scheme** and **Description** selection fields.

You use the → button to add selected available actions to the **Selected actions** field. If further settings are available for this action, EPLAN opens the corresponding settings dialog for further settings for this action after you move it.

For the **Back up data** action, for example, the Back up data dialog will be opened. Here you can set special settings for this script.

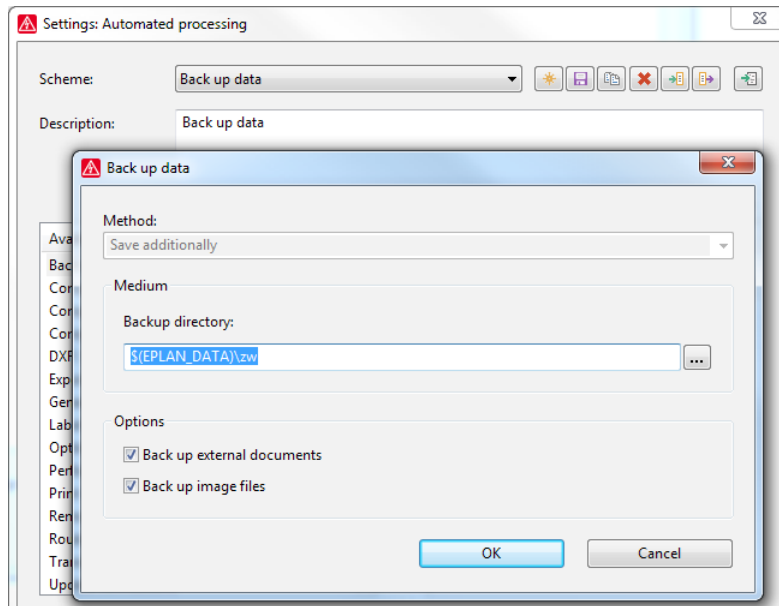


Fig. 9.31

Further dialog for data backup

If you now combine these settings, i.e. compress project followed by a data backup, then you can process it automatically.

Combining settings for scripts

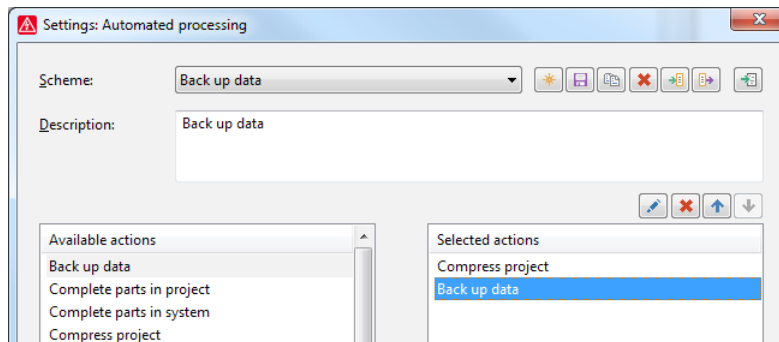


Fig. 9.32

Finished script with selected actions

This means that there are no more individual steps, you simply call up this one scheme and EPLAN fully automatically performs all the previously defined actions in a single step.

You can later edit the available actions via the top toolbar on the right. Select the corresponding action in the **Selected actions** field and open it with the *Edit* button.



NOTE: Created schemes are not stored on a project-specific basis. Using my customer code as an example, such a scheme would be located in the \Bernd Gischel\Scripts directory following the export.

In general, the last edited scheme is stored and can be started via the **UTILITIES/AUTOMATED PROCESSING** menu item. But you can also select and run other schemes here.

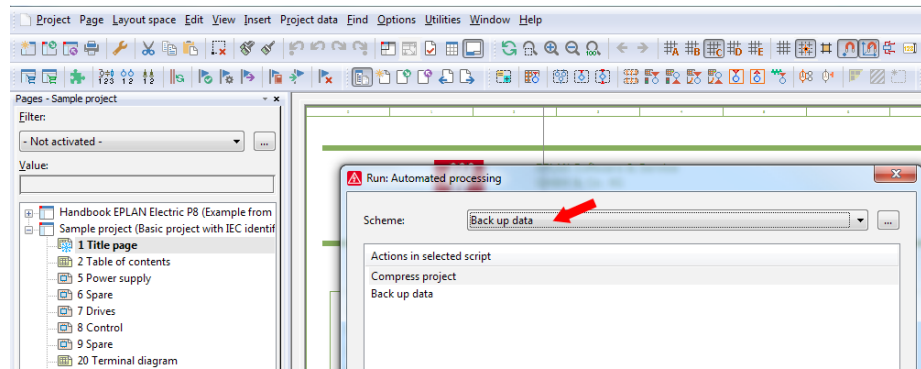


Fig. 9.33 Scheme selection

When the scheme is started, EPLAN executes the stored actions and displays the current progress in a dialog.

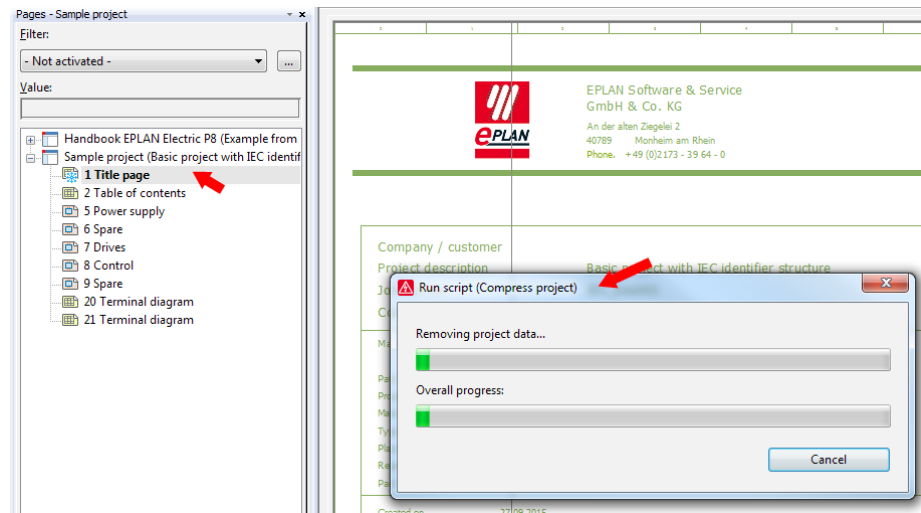


Fig. 9.34 Status of the script (scheme)

When the script (scheme) is finished, no other messages are displayed.

■ 9.4 Backing up and restoring master data

As well as projects, you can also back up and restore master data in EPLAN. This is a great advantage because it provides a unified, simple method of exchanging data between EPLAN users. EPLAN uses file extensions (among other features) to distinguish between the different types of data backups.

- **zw1:** File extension for projects
- **zw2:** File extension for symbol libraries
- **zw3:** File extension for plot frames
- **zw4:** File extension for forms
- **zw5:** File extension for macros
- **zw6:** File extension for parts data
- **zw7:** File extension for dictionaries
- **zw8:** File extension for settings not specific to the project (user and workstation data)

File extensions of
EPLAN backup files

9.4.1 Backing up master data

Backing up of master data generally always follows the same process. Master data is saved via the **PROJECT/BACKUP** menu item. During the backup process, you can select the master data type that should be saved.

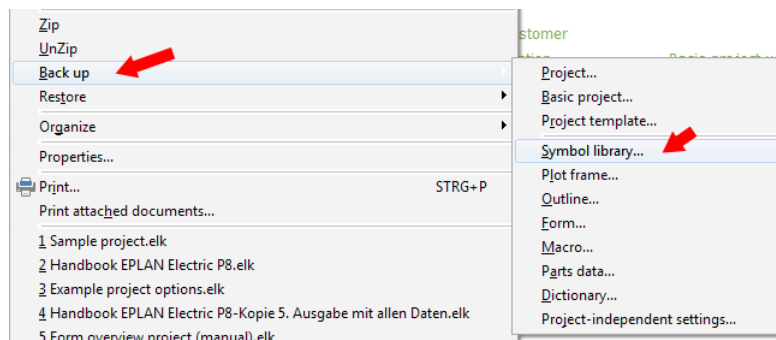


Fig. 9.35 Project/Backup/[Master data] menu

When master data is backed up via this menu item, EPLAN opens the corresponding **Back up [master data]** dialog, whereby **[master data]** represents the various types of master data such as symbol libraries, plot frames or the parts database. The title bar of the respective window also indicates which type of data backup dialog is currently being used.

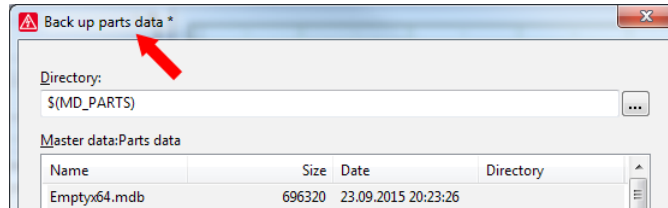


Fig. 9.36 Back up [Master data] dialog

The **Back up parts data** dialog here is used as a representative example, since all the dialogs have the same structure and are used in the same way.

A data backup dialog for master data has a window with an entry indicating which master data is to be backed up. The example illustrates backing up of parts data.

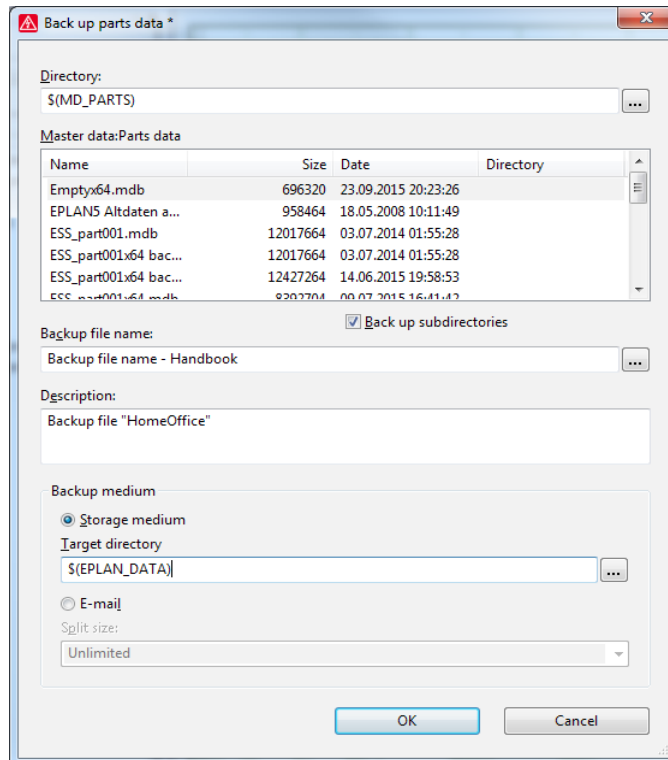


Fig. 9.37 Dialog as an example of how to back up master data

- **Directory:** A selection field defining the directory containing the master data to be backed up.
- **Backup file name:** An important entry since a backup cannot be performed without a name for the backup file. EPLAN uses this name as the file name for the backup file and appends the appropriate file extension.

- **Description field:** This field can (but does not have to) contain a description of the current data backup. There are no limits to the contents of this field. This description is later displayed in the *Description* field when restoring from the backup.

The fields/options such as **Storage medium** or **E-mail** have already been sufficiently described in the section on data backups for projects and will not be further discussed here.

When all settings are correct, you start the data backup process by clicking **OK**. EPLAN begins to back up the data and displays a final message after a successful backup. When the backup has been successful, a message will be displayed.

This completes the backing up of master data.

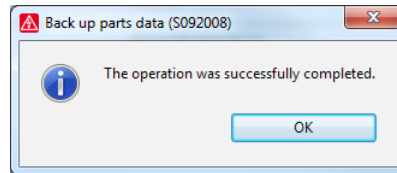


Fig. 9.38
Message confirming success

9.4.2 Restoring master data

You use the **PROJECTS/RESTORE [MASTER DATA]** menu to restore master data. As described in Chapter 8, “Export, import, print”, examples of master data are plot frames, forms, parts data and symbol libraries.

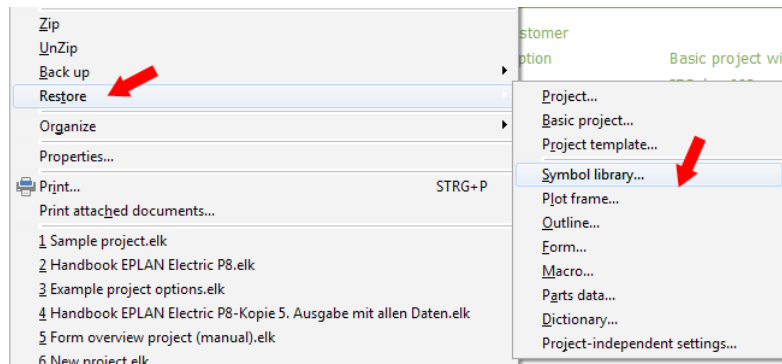


Fig. 9.39 Project/Restore menu

If you wish to restore this data in EPLAN, you use the **PROJECT/RESTORE [MASTER DATA]** menu to open that dialog.

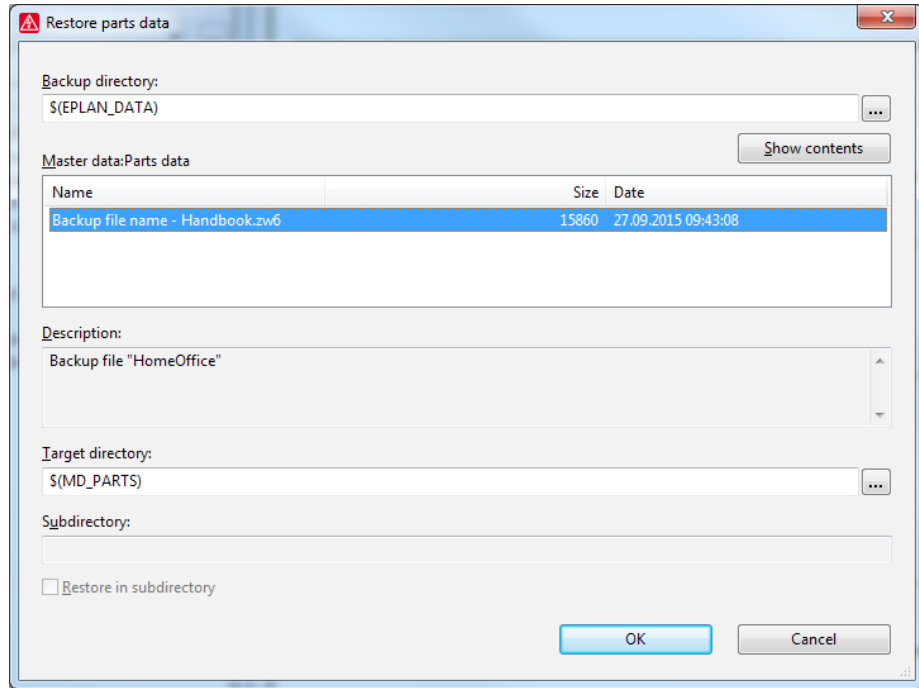


Fig. 9.40 Example of a Restore master data dialog

Again, EPLAN has standard dialogs for various types of master data, so that the dialog shown here – to restore master data – is identical to that of all other types of master data. You set the **Backup directory** to the directory containing the data you wish to restore. You can use the button to select the appropriate storage location if this differs from the default storage folder.

Once the correct folder is selected, the **Master data: [Master data type]** field lists the available backups (files).



NOTE: You can only restore one file here. Multiple selection is not possible.

The **Description** field displays the information entered when the master data was backed up. This field is purely informative.

To restore the master data to a different specific directory, you need to set up a suitable folder that differs from the default directory in the **Target directory** field. The default directory is the original source directory of the master data, i.e. the location where the backup was performed.

Once all entries match your requirements, you start to restore master data by clicking **OK**.

EPLAN restores the selected master data back to the specified directory and then displays a message with information about the process of restoring the master data (or about the failure of the process).

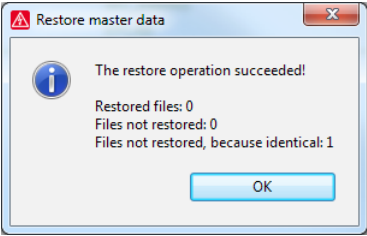


Fig. 9.41
Message confirming success

9.5 Send project by e-mail directly

In addition to sending an e-mail as part of a normal backup process, in EPLAN it also possible to send a project directly via e-mail. You can access this function directly via the PROJECT/SEND BY E-MAIL menu item.

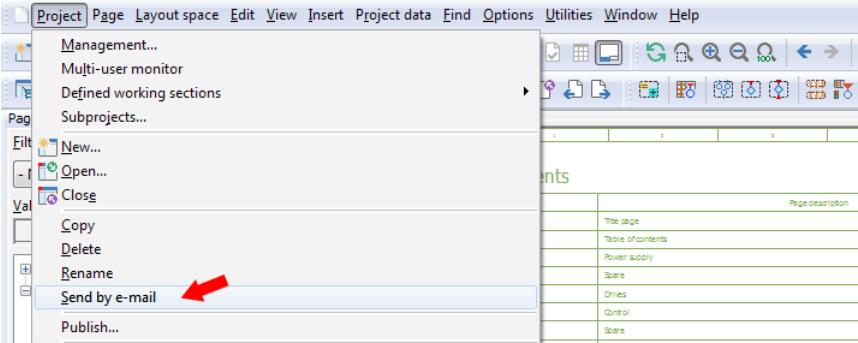


Fig. 9.42
Send project by e-mail directly

You select the desired project in the **page navigator** and then call up the **Send by e-mail** function. EPLAN displays a prompt. Then, EPLAN handles the default settings for sending by e-mail (e.g. the stated size of split files) and begins to generate the backup files.

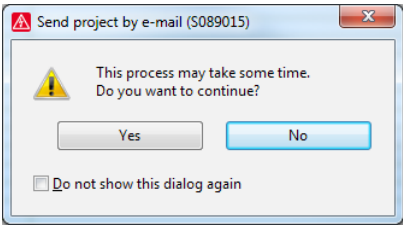


Fig. 9.43
First note regarding e-mail sending

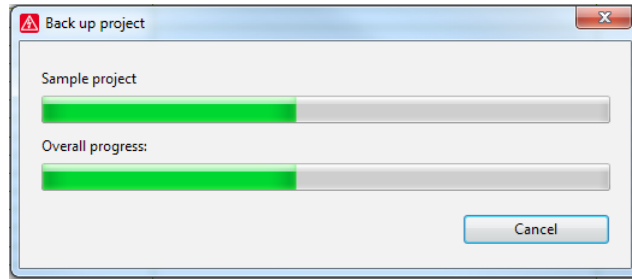


Fig. 9.44 Progress dialog for backing up project by e-mail

Depending on the size of the files, EPLAN now generates one or several e-mails with the split project files and then waits for the user to send these e-mails.

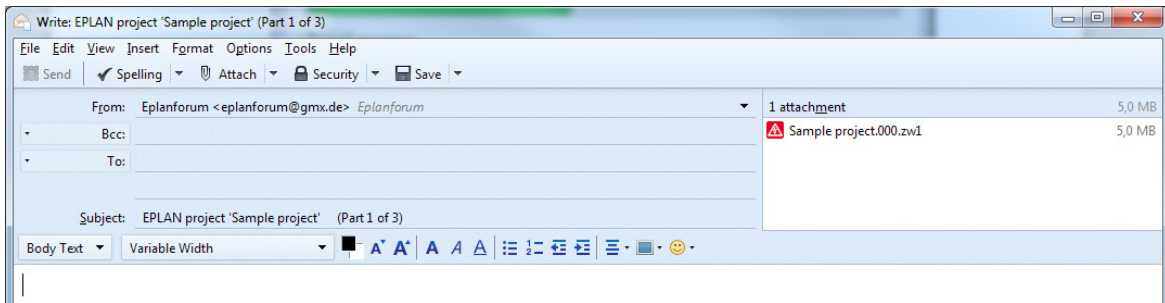


Fig. 9.45 Newly generated e-mail with the split project files

When all e-mails are sent, a message stating that they were successfully sent is displayed. The complete project has now been sent.

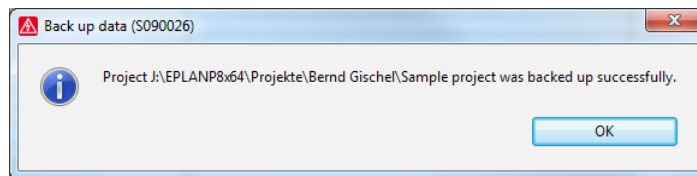


Fig. 9.46 Message that e-mails have been successfully sent



NOTE: You can send only one project at a time by e-mail from the page navigator. Multiple selection, i.e. sending several projects by e-mail at once, is not possible in the page navigator.

10

Master data editors

EPLAN supplies a large number of finished system master data for forms, plot frames and also symbols. These are normally sufficient for most applications. However, if special conditions for appearance exist or if particular project or device data is to appear in the forms and this is not possible with the standard EPLAN reports (forms) supplied, then some new system master data such as reports (forms) will need to be created from scratch.



NOTE: In EPLAN, forms are usually used for reports. This is why we can speak of reports or forms. But not all reports are forms. A report can also be an Excel table that was created, for example, with the *Labeling* function or *Edit properties externally*.

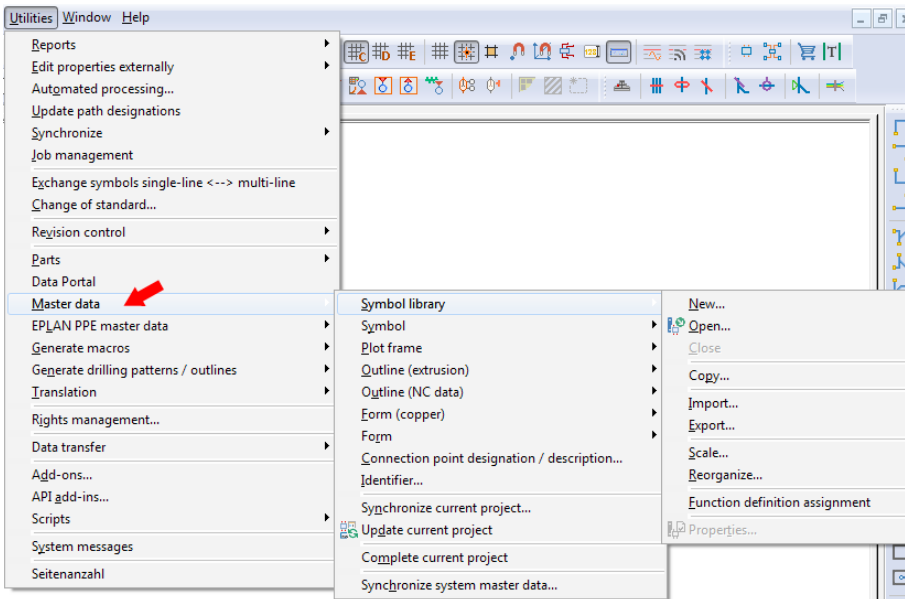


Fig. 10.1
The master data menu
with the possible
editors



NOTE: EPLAN always opens master data for editing on a temporary basis in the current project. It does not matter which project is involved. But a project must be open, otherwise master data cannot be selected for editing.

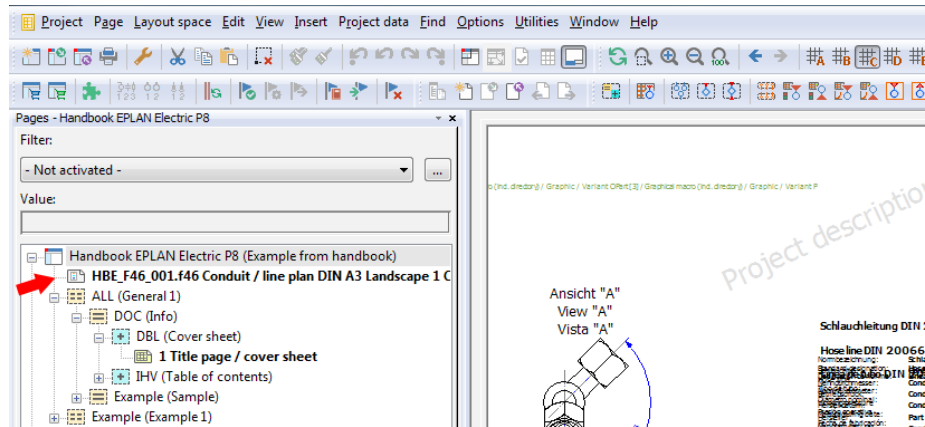


Fig. 10.2 Example representation for master data opened for editing

Generally, it is always system master data that is edited. Master data editors never modify project master data. It is rather the form or plot frame in the system master data directories that is modified and/or created and then saved in the system master data directories.

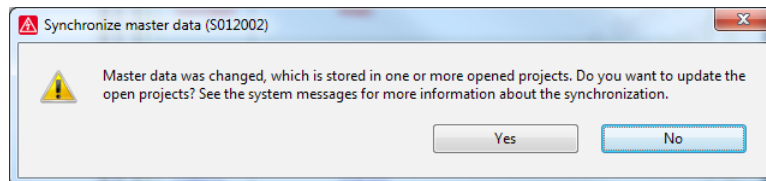


Fig. 10.3 Prompt when stored master data has been modified

If modified master data is to be used in the current project, then after it is modified the system master data must be synchronized with the project master data. In other words, the automatic prompt whether data is to be synchronized must be answered with YES. Otherwise, the modifications to the master data will not be used in the current project, because it has not been updated.

You access the various editors via the menu **UTILITIES/MASTER DATA [TYPE OF MASTER DATA]**. The **[FUNCTION]** can be symbol libraries in general, symbols, plot frames or forms.

NOTE: Editing outlines and forms will not be covered in this version of the manual.

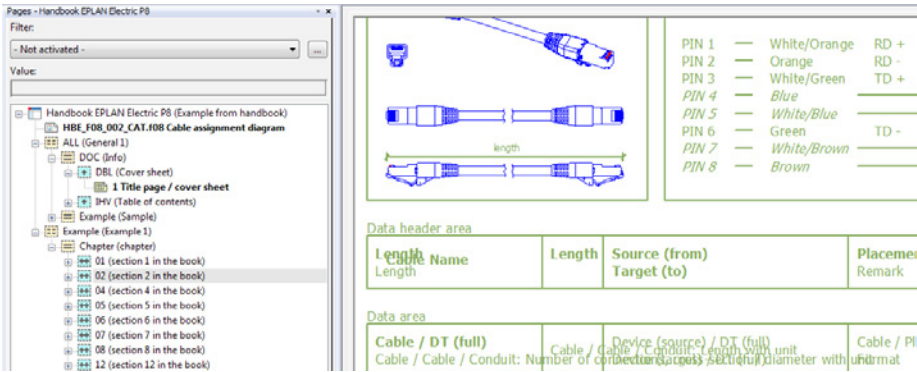


Fig. 10.4 Example of form structure

NOTE: Before you start editing master data, you should keep the following in mind: Original master data supplied with EPLAN should never be modified. It is possible to modify this data, but if EPLAN master data is a good basis for your own master data then I recommended that you copy it and assign a new name. You can then modify this copy as you wish.

If you follow this recommendation, then, when performing updates etc., the new, possibly modified original EPLAN master data in your system data directories can simply be replaced with the EPLAN master data of the same name. This means that you can benefit from improvements or extended functions immediately, and you will not have to integrate them into your own master data.

TIP: It is fastest and easiest to use existing master data as a basis and then edit it. You can create the data from scratch. But this will take more time and requires a more thorough understanding of the master data editor.

This chapter provides a brief explanation of the basic function steps required for editing and understanding. Not all the details and possibilities of the form editor, plot frame editor, symbol editor or symbol library editor can be explained here. These editors are simply too complex for this because there is a (practically) infinite number of options for implementing various requirements in connection with the master data.

■ 10.1 Preparatory measures

Although it is convenient to edit a project's system master data, this can also be “dangerous”. The more you edit, the greater the “danger” of losing track of what you have done: What forms/plot frames have I already created, what do they look like, and which ones can I use as templates for a new form/plot frame?

The upshot is that you might have to click through many forms/plot frames, and you might not be able to see all the information you need on the small preview. You load the form, only to discover that it is not the one you used last week in another project.



TIP: If a multiple screen solution is used, it is best to enable the separate graphical preview. The window size of the preview can be modified, and you can view the forms/plot frames in a convenient size. The graphical preview is activated in the View/Graphical preview menu.

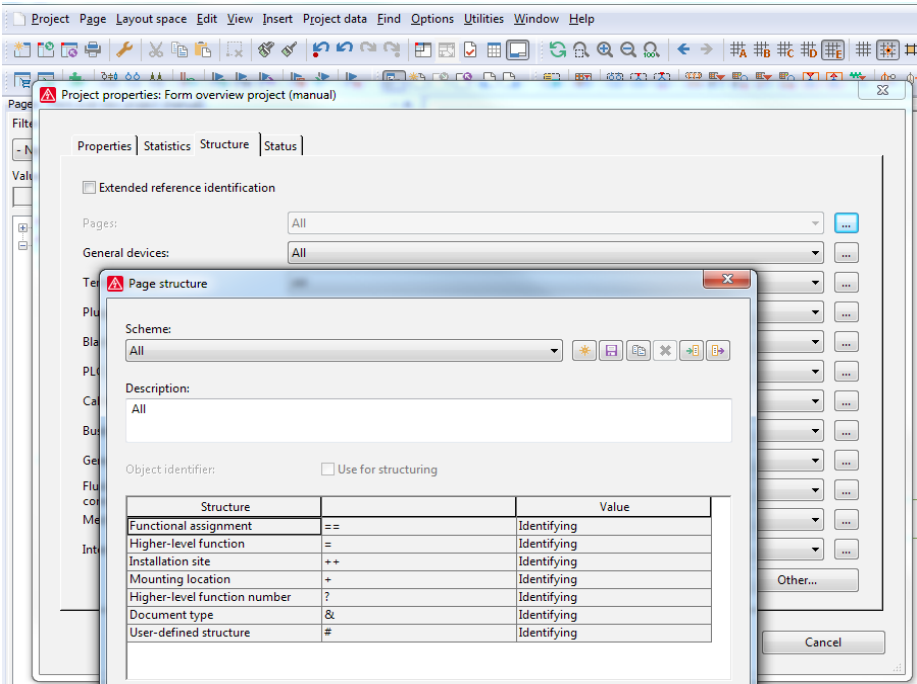
■ 10.2 Clear overview of forms

To help prevent the onset of “master data chaos”, you have various options at your disposal to display forms/plot frames in a structured manner.

10.2.1 First option — manual overview

A form overview would be a type of “form project”, which is created and maintained manually. But it is really nothing other than a normal schematic project. Depending on your personal wishes and organization, you can store all forms and plot frames as extra pages in this project. This means all forms/plot frames are gathered in a project and created as separate pages, and the corresponding form or plot frame is assigned to such a page.

The selected page structure covers all identifiers, which allows deep structuring of the pages by customer and page type (in this case report type). However, this is a matter of personal taste and users can decide for themselves.



Possible structure of a project providing an overview of all forms and plot frames

Fig. 10.5
A possible page structure

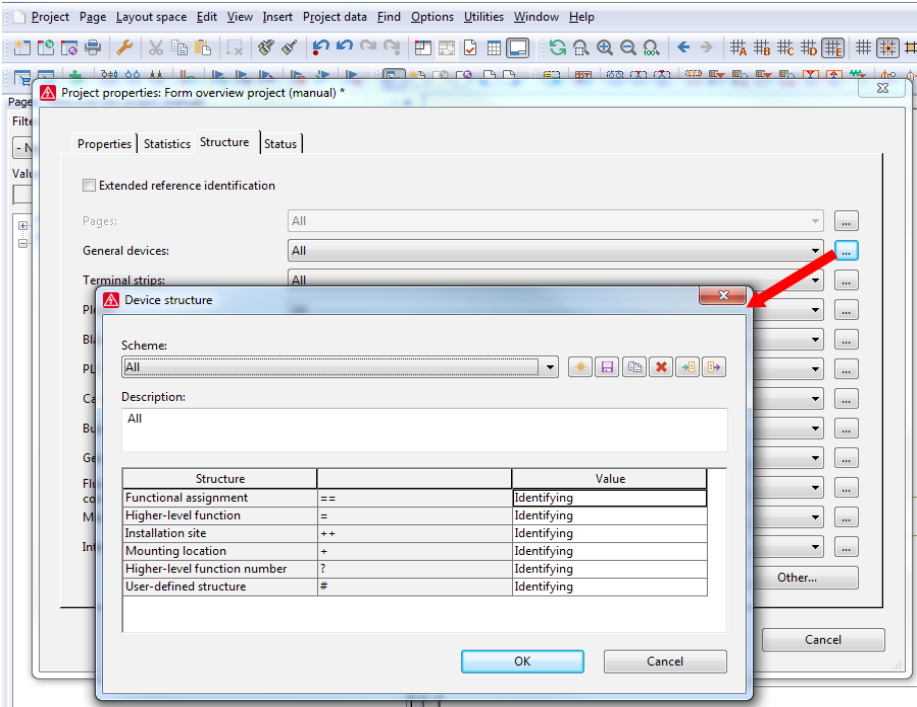


Fig. 10.6
Structuring of the remaining objects

Of course, you cannot directly edit the form or plot frame on these pages. We do not want to do this anyway. When the project is always maintained with the master data, you have a good overview of which forms/plot frames have been created over time, and, if needed, you can select a particular form or plot frame for subsequent editing.

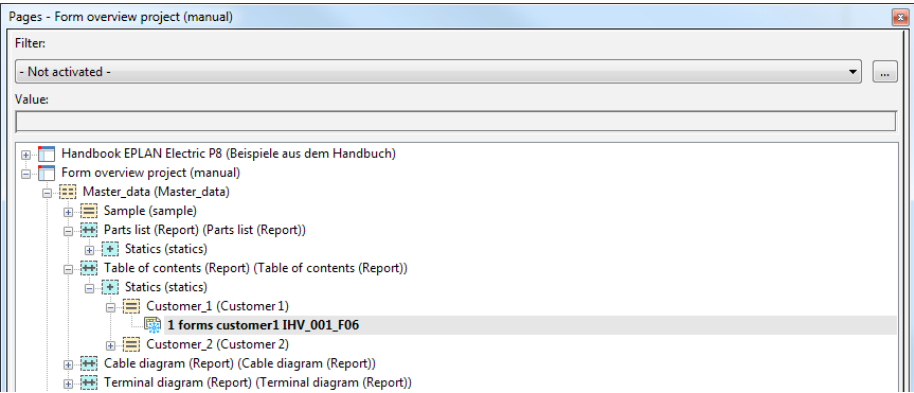


Fig. 10.7 Possible layout of the page structure with the various types of identifiers

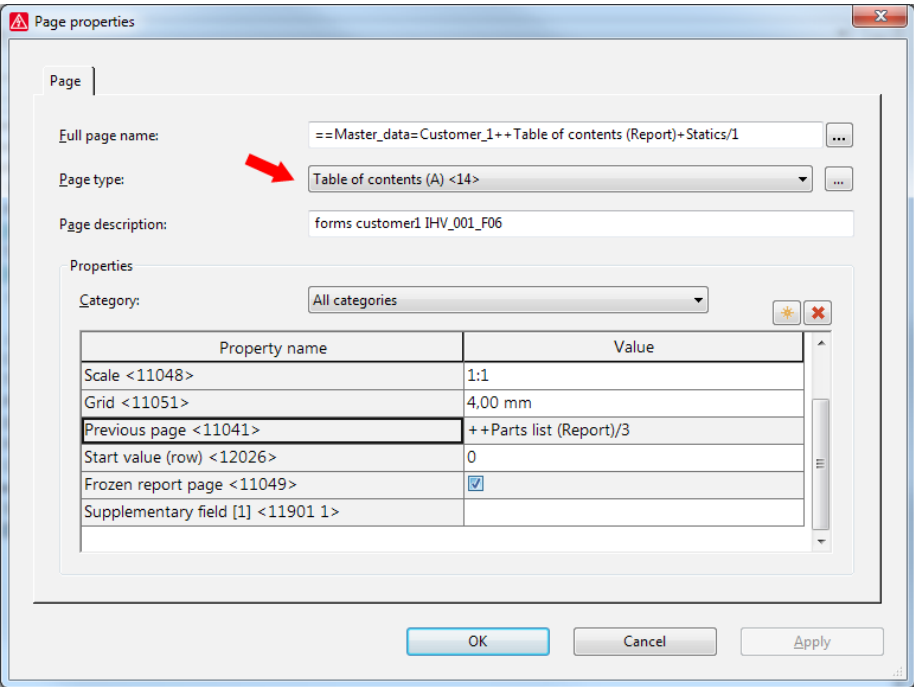


Fig. 10.8 Page properties of a page in the project

The relevant page types (e.g. here a cable diagram) are set for the pages, the page descriptions are filled, and the form is selected via the selection in the `<11015 Form name>` property. This stores it in the page and displays it.

10.2.2 Second option – automatic overview

A second option for documenting forms and plot frames is to use EPLAN's own media to create an overview.

In this case you also create a new project and store all master data (forms, plot frames) in the project from the corresponding customer system master data directory. The easiest way to do this is via Synchronize master data (UTILITIES/MASTER DATA/SYNCHRONIZE CURRENT PROJECT menu).

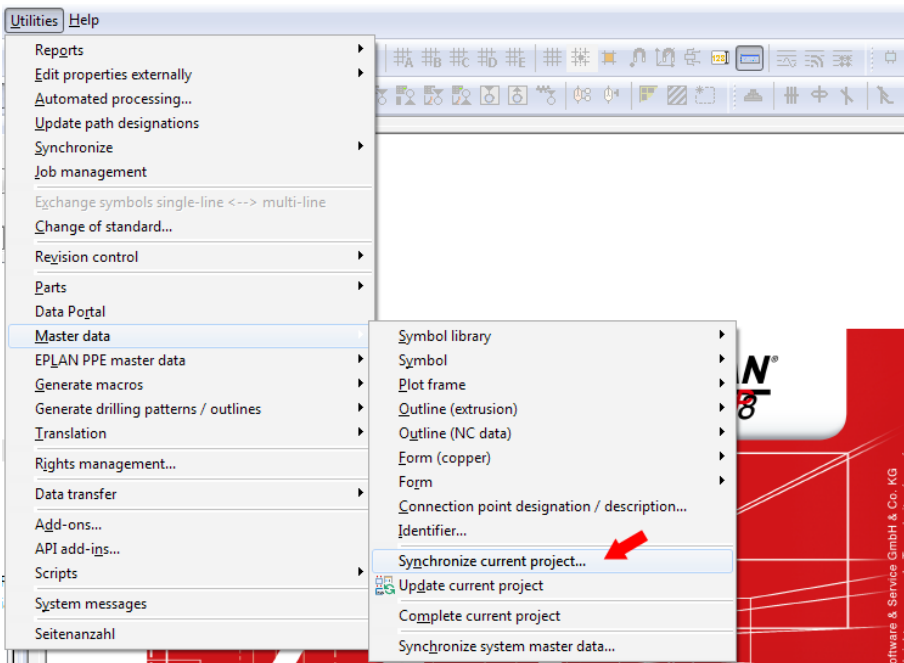


Fig. 10.9 Synchronize master data menu for a project

In the subsequent **Synchronize master data** dialog, you can then copy to the project all (non-)existing system master data. **Note:** No filter may be active.

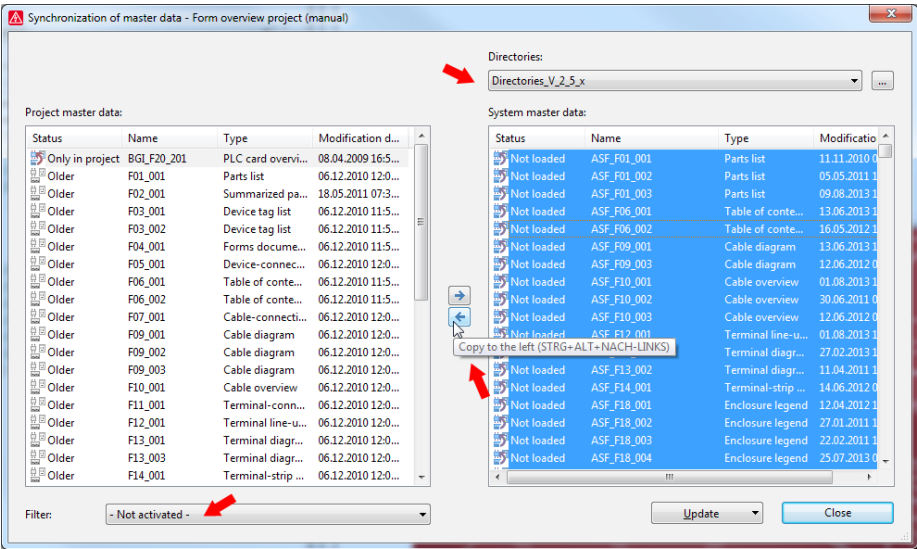


Fig. 10.10 Synchronize master data dialog

If all master data like forms and plot frames have been copied to the project, you can then generate the *Forms documentation* and *Plot frame documentation* reports. EPLAN automatically generates the corresponding project overview for all forms and plot frames in the project. You can also use this as an overview of the existing master data in the system directories.

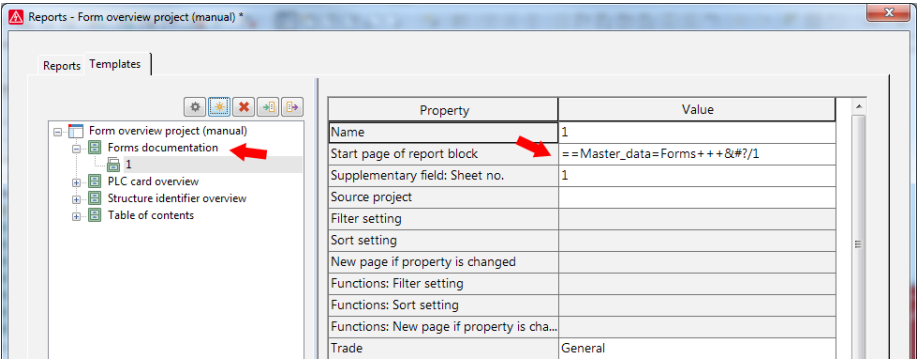


Fig. 10.11 Templates for the forms and plot frame documentation reports

A possible final result is shown in Fig. 10.12.

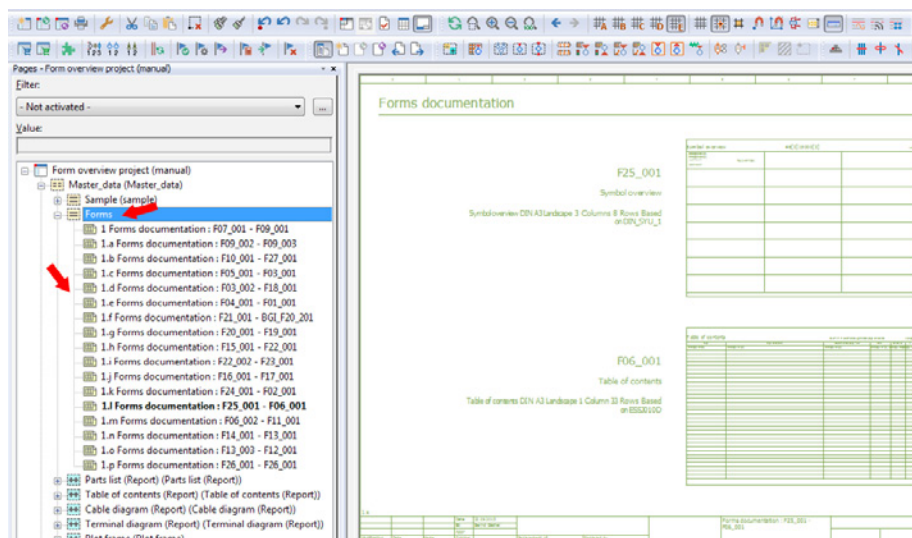


Fig. 10.12 Representation in the page navigator

Unfortunately, there are no other sorting options for forms documentation or plot frame documentation. But it is sufficient for a quick overview.

The great advantage of automatic generation of forms documentation or plot frame documentation is clearly that the relevant pages are generated automatically by EPLAN, with literally with just a “click”. You can’t get any faster than that!

10.3 Forms

Forms are used to graphically insert evaluated project data into the project documentation.. Examples of forms are terminal diagrams, cable diagrams, device tag lists or a table of contents.

The supplied master data does not always exactly match the existing project data. This may be due to page references that are too long, function texts that do not fit in the intended rows and columns, or the need to “spruce up” the display of PE terminals with a small graphical symbol. Consequently, reports will need to be adjusted or created from scratch.

Menu for editing forms

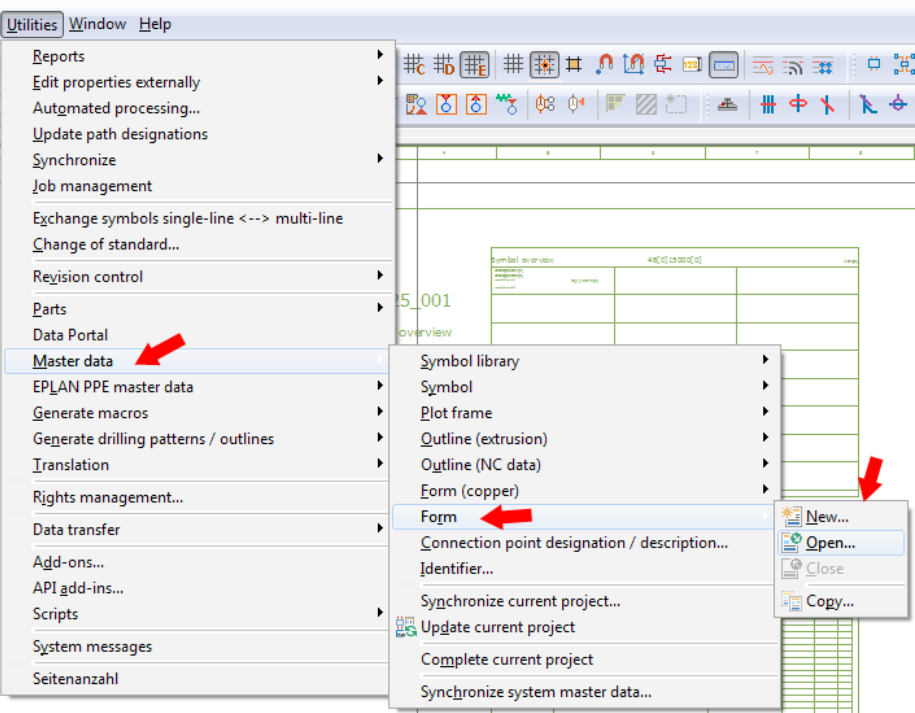


Fig. 10.13 Calling the form editor

EPLAN provides a number of different ways of adjusting a form to the new layout. Forms can be created from scratch, edited (i.e. opening and changing existing forms) or copied and then modified.

The form structure can be accessed via the **UTILITIES/MASTER DATA/FORM** menu.

NOTE: The form editor does not have a **Save** function. If changes are made and the form editor is exited via the **CLOSE** function in the **UTILITIES/MASTER DATA/FORM** menu, then the form is automatically stored with these changes in the system master data.

The project master data, however, still contains the previous “old” form, which you could, if necessary, copy back to the system master data directory. This means that, as long as no (manual) synchronization of the master data (system master data with the project master data) is performed (e.g. via the **UTILITIES/MASTER DATA/SYNCHRONIZE CURRENT PROJECT** menu), then this method can be used to return the edited form in the system master data back to the state of the project master data.

After starting the function, EPLAN opens the **SYNCHRONIZE MASTER DATA [PROJECT NAME]** dialog. You now select the form in the project master data area and use the  button to copy it into the system master data area.

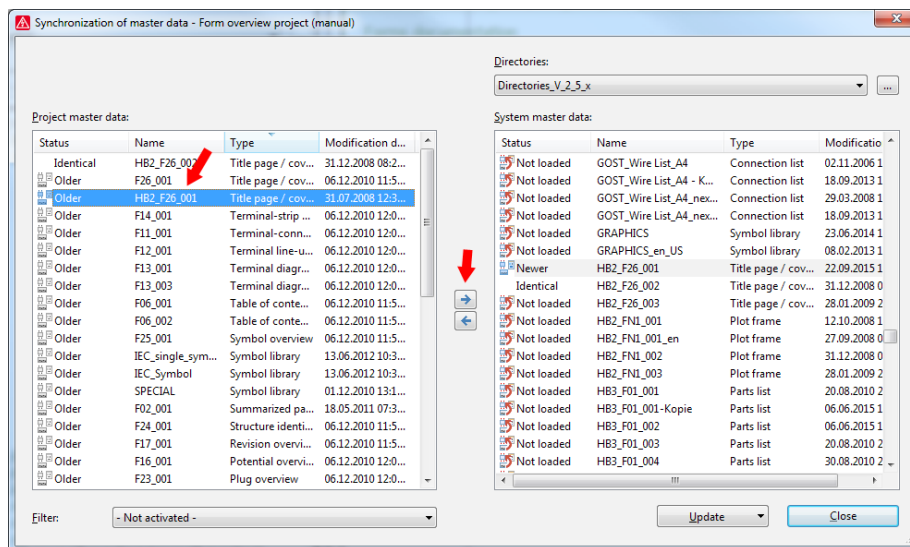


Fig. 10.14 Synchronize master data dialog

When existing master data is found, EPLAN opens the **COMPARE MASTER DATA** dialog. A decision must be made here for each master data file as to whether the system master data should be replaced with the project master data.

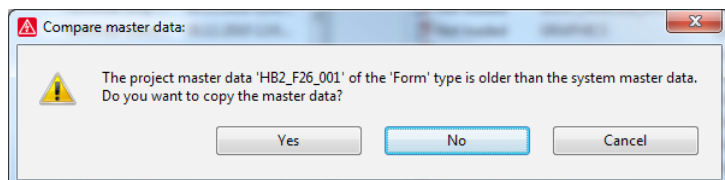


Fig. 10.15 Prompt asking whether the system master data should be overwritten

If you confirm the dialog with **YES**, the system master data is overwritten with the project master data and EPLAN displays a notification message that the system master data was updated. In addition, the system messages contain further information about the master data synchronization just completed.

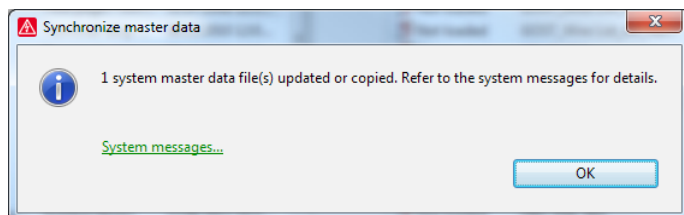


Fig. 10.16 Result message in the Synchronize master data dialog

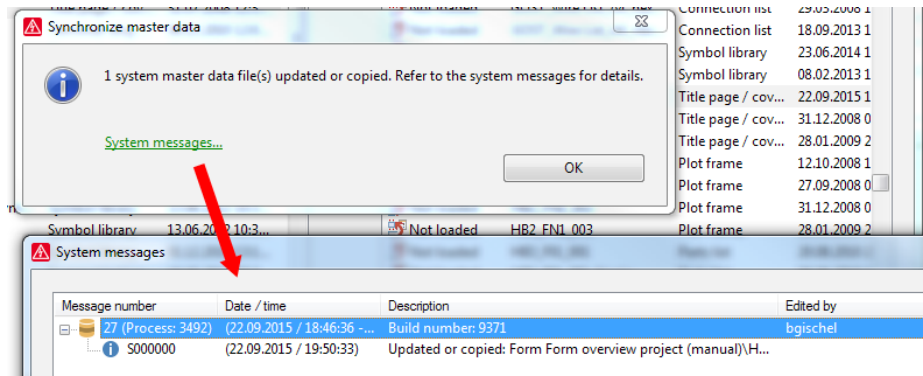


Fig. 10.17 System messages

 **NOTE:** This method described is really only an “emergency plan” for restoring master data that was unintentionally changed. As already mentioned, EPLAN allows automatic synchronization of master data when projects are opened.

If master data is incorrectly reset to an old state, unwanted master data might be synchronized in existing projects (if the automatic master data synchronization has been activated). You should therefore be extremely careful when using this feature.

10.3.1 Create new form (from copy)

The easiest method to create a form is based on an existing one. It makes no difference what type of page is currently open in the project. You call up the **UTILITIES/MASTER DATA/FORM/COPY** menu and EPLAN opens the **Selection/Copy form** dialog.

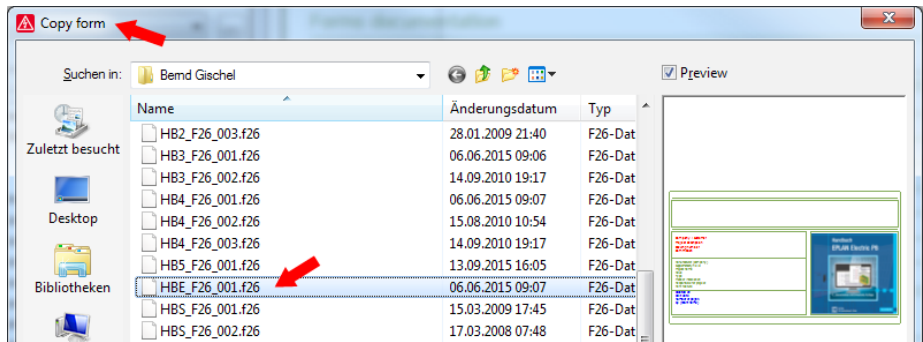


Fig. 10.18 Copy form dialog

In this dialog, you select the form to be copied. Confirm the selection by clicking the **OPEN** button in the lower area.

EPLAN then opens the **Create form** dialog. Here you enter the new form name in the **File name** field.

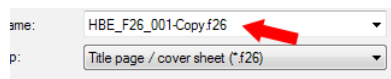


Fig. 10.19

Entry in the File name field

The form can now be saved via the **SAVE** button. EPLAN closes the dialog, opens a temporary page with the selected **[Form] page type** and displays the copied form on this page for editing.

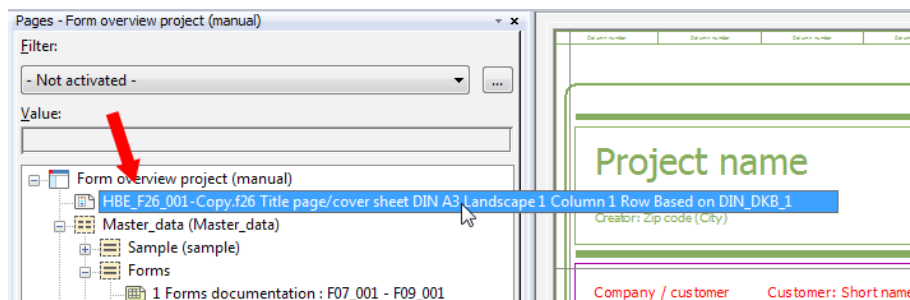


Fig. 10.20 Temporary page with the form in the page navigator

The form can now be edited. In the form editor, all graphical functions such as **Insert line** or **Insert circle** (via the **INSERT/GRAPHIC** menu) are available, as well as functions for inserting normal text.

The **INSERT** menu contains a number of other functions relating only to the form editor. The graphical functions have already been explained or are well-known functions in the graphical editor.

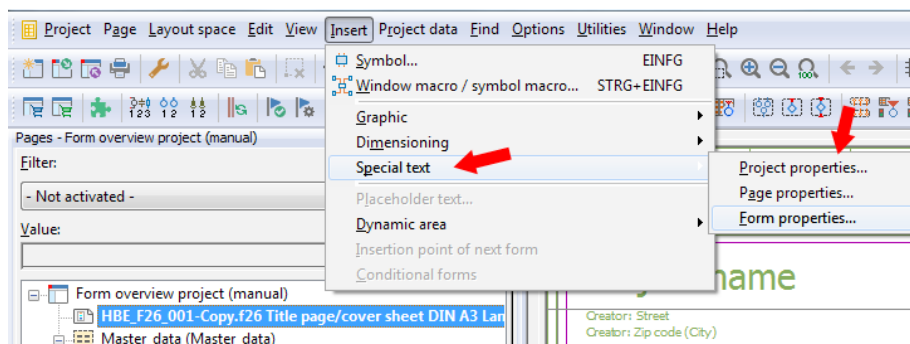


Fig. 10.21 Special texts

Special texts such as project, page and form properties can be placed in forms. The actual form texts (data for reports) are, however, called “placeholder text” in EPLAN. In contrast to the project and page-specific texts or form properties, these placeholder texts are only filled with project data after the reports are graphically generated.

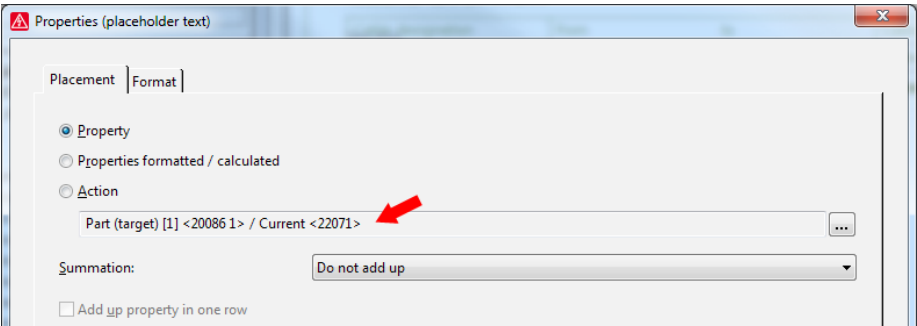


Fig. 10.22 Display placeholder text dialog

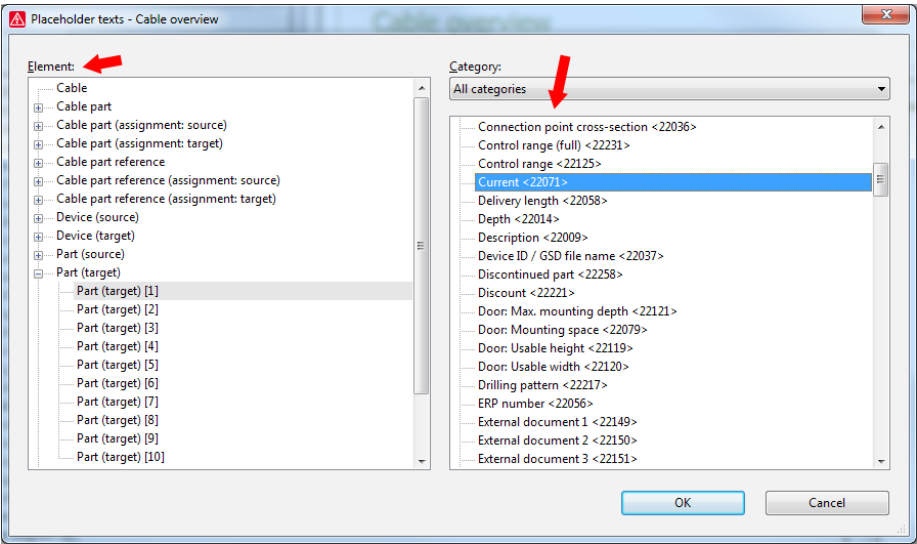


Fig. 10.23 Select placeholder text dialog

Insert placeholder texts

Open the **INSERT/PLACEHOLDER TEXT** menu, and in the **Placeholder text** dialog on the **Placement** tab, use the **...** button in the subsequent **Placeholder texts – [Report type]** dialog to select a placeholder text.

The report shown here is an example – other placeholder texts apply to all other forms. In our example, the element in the left area is preselected (left selection area in the dialog), as is the desired property that is required in the right area of the dialog. You can now confirm this selection by clicking **OK**.

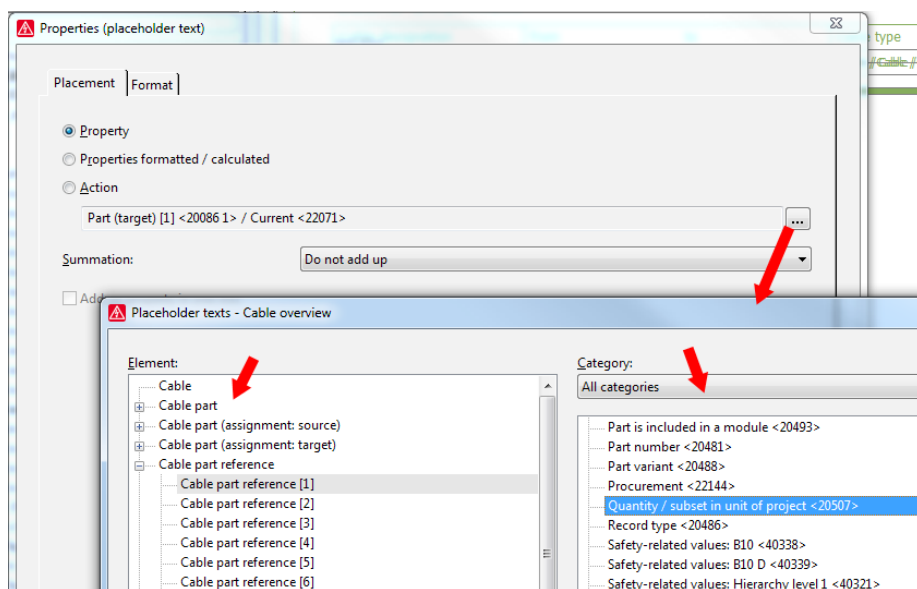


Fig. 10.24 Preselection (left) and selection (right) of the property

EPLAN closes the dialog and transfers the selected placeholder text to the Placeholder text dialog.

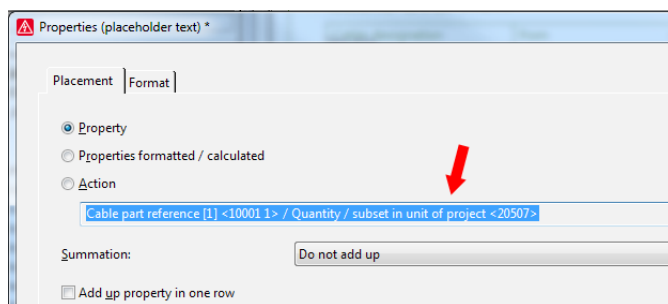


Fig. 10.25 Placeholder text applied

Confirm the **Placeholder text** dialog with OK. The placeholder text now hangs on the cursor and can be placed at any (practical) position in the form.

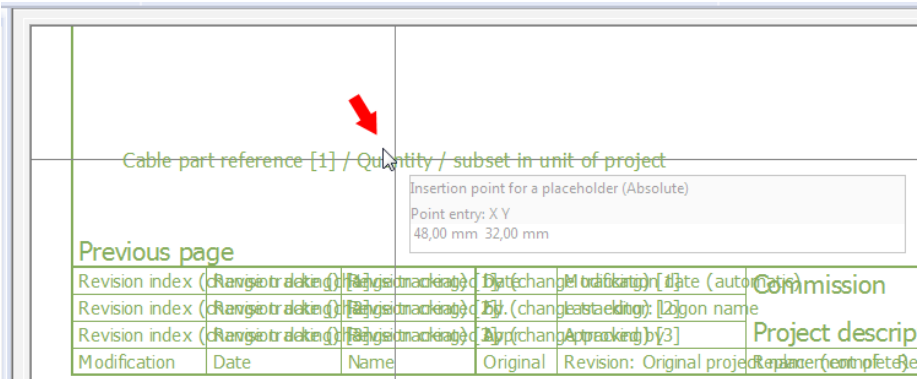


Fig. 10.26 Selecting the placement

As with any other (free) text, placeholder text can be freely formatted. This allows the form to be appropriately “constructed”. It is also possible to subsequently modify placeholder text, i.e. select a different one. You select the placeholder text and display its **properties** with a double-click or by clicking the popup menu.

You then proceed as described above. Use the  button to call up the subsequent dialogs and select, apply and place other properties in the usual manner.

When you are done, the form can be checked. Using the function in the UTILITIES/CHECK FORM menu, EPLAN can check the form for errors. If the form is OK, the following message is displayed: “The form is OK.”

Check form function

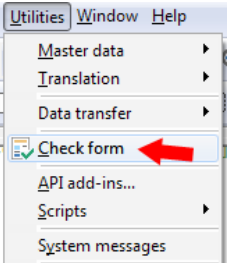


Fig. 10.27

Check form

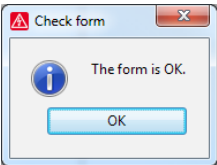


Fig. 10.28 Message that the form is OK

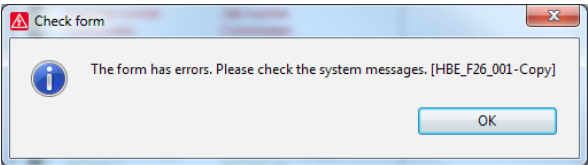


Fig. 10.29 Error in the form

If the form has errors, EPLAN indicates this and writes the errors to the system messages, but also indicates the error already in the subsequent dialog.

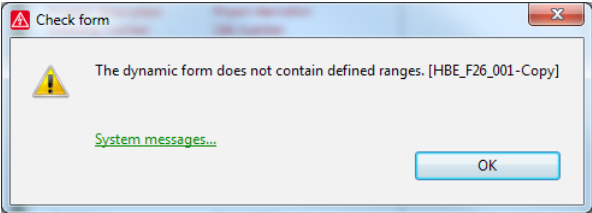


Fig. 10.30

Detailed error information

These should be corrected to avoid incorrect system master data.

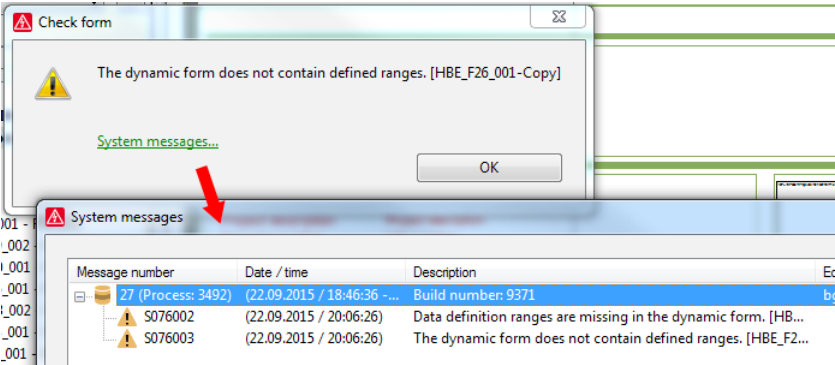


Fig. 10.31 System messages about the form errors

When handling forms, EPLAN distinguishes between dynamic forms, where only certain areas exist (e.g. a data area that contains an X amount of data), and static forms, where the amount of data is permanently “wired” into the form.

Dynamic or static form handling

Dynamic forms expand their graphical display area depending on the amount of project data. The maximum amount of data that a dynamic form should display is defined in the form properties. In our example (the procedure is the same for all forms), you can access the form properties via the key combination **CTRL + M + D** or in the page navigator via the **PROPERTIES** function in the right-click popup menu.

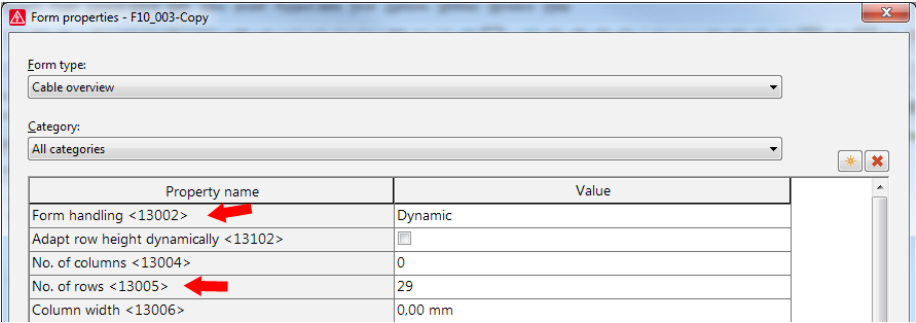


Fig. 10.32 Form properties dialog

EPLAN opens the **Form properties** dialog. All relevant form data is defined here. How should the form be handled (dynamically or statically), which entries in the *automatic page description* property should be written when project data is evaluated, or what is the maximum amount of data (number of rows) that the form should display before a page break or new report page is created?

Form properties

An example of a dynamic form is shown in Fig. 10.33. The rows are dynamically extended depending on the volume of data. If a terminal strip has five terminals, then the form will contain five rows with data. A terminal strip with twelve terminals will show twelve (data) rows in a report.

Column number	Column number	Column number
Cable overview		
Cable designation	from	to
Cable / DT (full)	Device (source) / DT (full)	Device (target) / DT (full)
Data footer area		

Fig. 10.33 Example of a dynamic form

In principle, static forms have the same properties as dynamic forms. In static forms, the data areas are fixed, i.e. permanently defined, and the maximum number of data rows is entered in the form properties – in the same way that the graphic for the form was created with the maximum number of data records.

An example of a static form is shown in Fig. 10.34. The difference can be clearly seen. Whereas a dynamic form has a data row that is dynamically expanded, a static form already has all data rows as a finished graphic. The data is written to these graphical rows when generating a report.

Column number	Column number	Column number	Column number	Column number
Cable overview				
Cable name	Source (from)	Target (to)	Cable type	all conductors
Cable / DT (full)	Device (source) / DT (full)	Device (target) / DT (full)	Cable / Cable / Conduit: Type	Cable / Cable / Conduit: Number / DT (full)

Fig. 10.34 Example of a static form

If the form is OK, then you can exit the form via the UTILITIES/MASTER DATA/FORM/CLOSE menu. It is then permanently “saved”.

NOTE: After the very first editing, a copied form is automatically stored in the system master data and (when used) also stored in the project master data. At this point, synchronization of the master data (system and project) does not need to be performed.

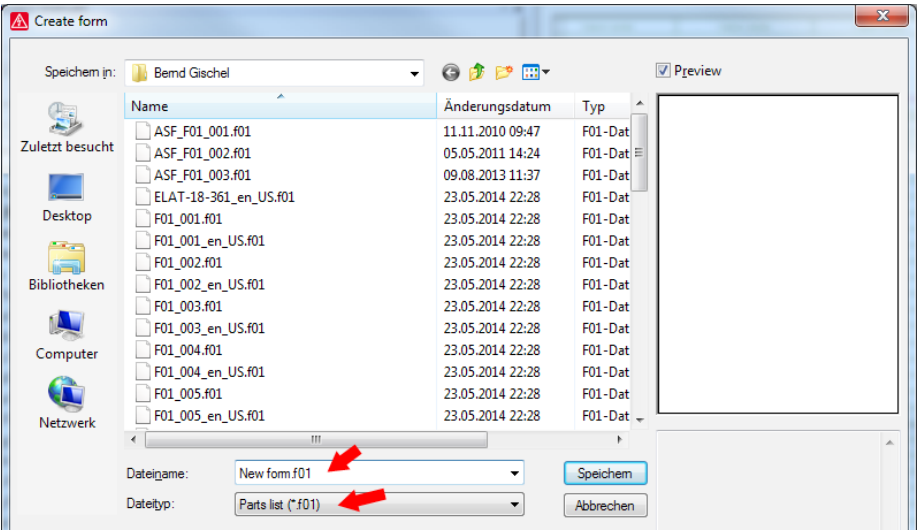


Fig. 10.36
Defining form name

The **Create form** dialog can now be exited via the **SAVE** button. EPLAN opens, in the now familiar manner, a temporary page in the background and the **Form properties** dialog is displayed in the foreground. You can already begin making the first entries here. However, this is not absolutely necessary at this time and you can enter this data later.

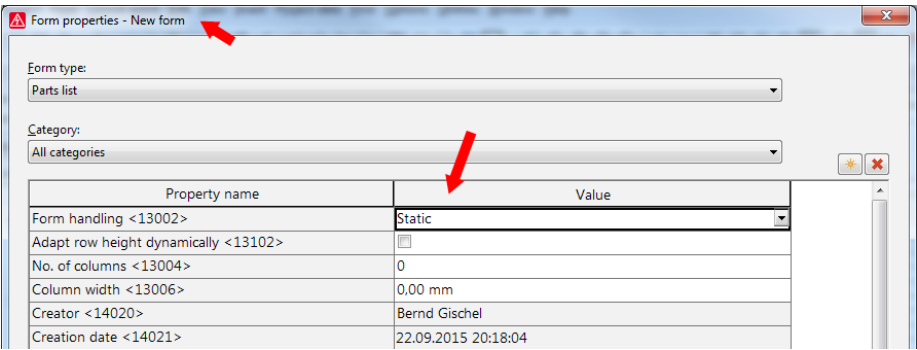


Fig. 10.37
Form properties dialog

As described in the previous sections, form editing is done using the various different functions, such as inserting graphical elements or texts.

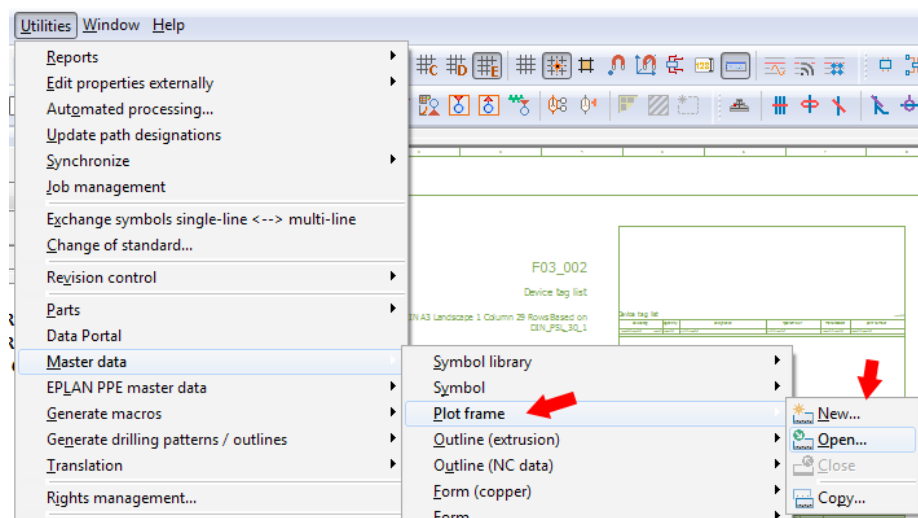
Creating a new form can be a time-consuming process because you start with “blank” form. It is therefore recommended that you copy a similar form and then modify this copy.

NOTE: The procedure for creating and editing a new form is identical to the procedure for editing existing or copied forms and is therefore not explained further.

10.4 Plot frames

Plot frames are a sort of frame around the actual project pages. They contain, for example, structure identifiers, column and/or row information and other data like page description, etc.

Editing plot frames in the plot frame editor is very similar to editing forms in the form editor. The editor is accessed via the **UTILITIES/MASTER DATA/PLOT FRAME** menu.



Menu for editing plot frames

Fig. 10.38 Calling the plot frame editor

As with the form editor, this menu has the items **NEW**, **OPEN**, **CLOSE**, and **COPY**. This menu functions in a similar way to the menu used when editing forms. We will therefore only deal with the features that are special to the plot frame editor.



NOTE: The plot frame editor does not have a Save function either. To restore plot frames that were accidentally modified, you should therefore use the Undo function in the plot frame editor or consider synchronizing existing project master data with the system master data.

10.4.1 Create new plot frame (from copy)

To copy and then edit a plot frame, in the **UTILITIES/MASTER DATA/PLOT FRAME** menu, select the **COPY** function. EPLAN opens the **Copy plot frame** dialog.

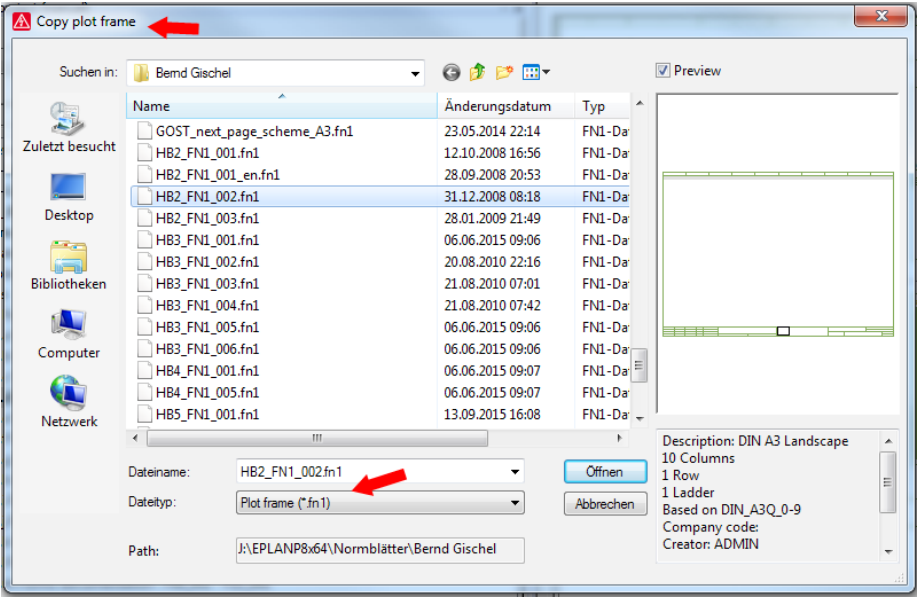
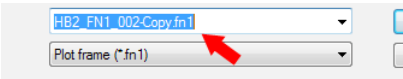


Fig. 10.39 Copy plot frame dialog

Fig. 10.40
Create plot frame
dialog

In this dialog, you select a suitable plot frame and confirm your selection by clicking the **OPEN** button in the lower area. EPLAN closes the **Copy plot frame** dialog and opens the **Create plot frame** dialog.



Enter the new name of the plot frame into the **File name** field. When you click **SAVE** to close the dialog, EPLAN temporarily opens (in the current project) a page with the copied plot frame.

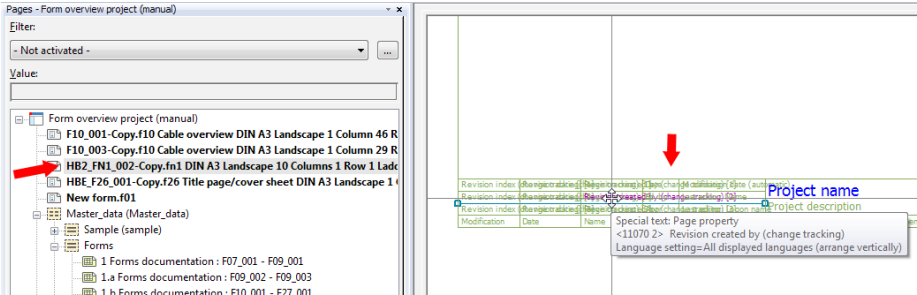
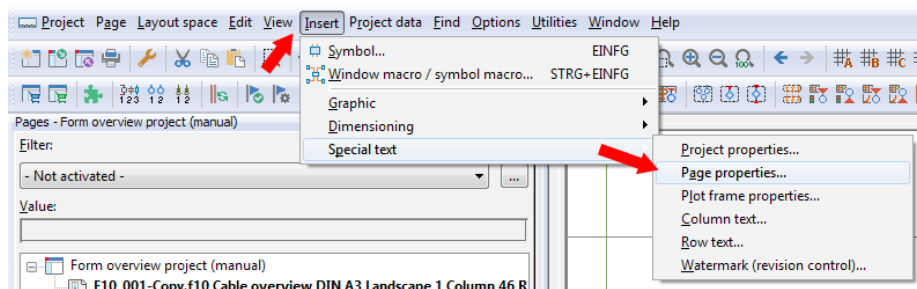


Fig. 10.41 Opened plot frame in the page navigator

The plot frame can now be edited using the same functions as already described in the form editing section 10.3. Graphical elements and texts can also be placed in the plot frame editor.

In contrast to forms, it is possible (and, in part, for project planning necessary) to specify column text (paths) and row text for the plot frame. You can also embed a special text watermark in the plot frame. This special text is then automatically filled with data from revision management.



Special texts for plot frames

Fig. 10.42 Insert/Special text menu

Column and row texts are treated as normal text. This also includes the formatting (font size, color, etc.).

Column and row texts are placeholder texts that are defined by entries in the **Plot frame properties** dialog (via the CTRL + M + D shortcut key or in the **page navigator** by selecting the temporary page, opening the popup menu, and selecting the **PROPERTIES** item) and that are later automatically filled in the project.

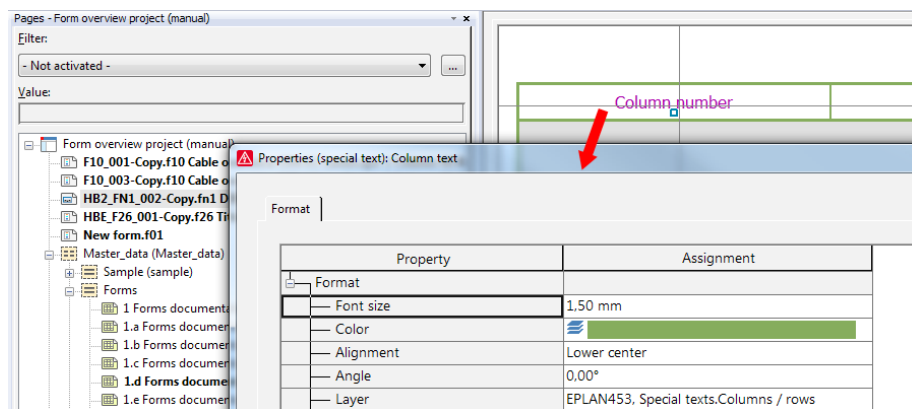


Fig. 10.43 Formatting options for the column and/or row texts

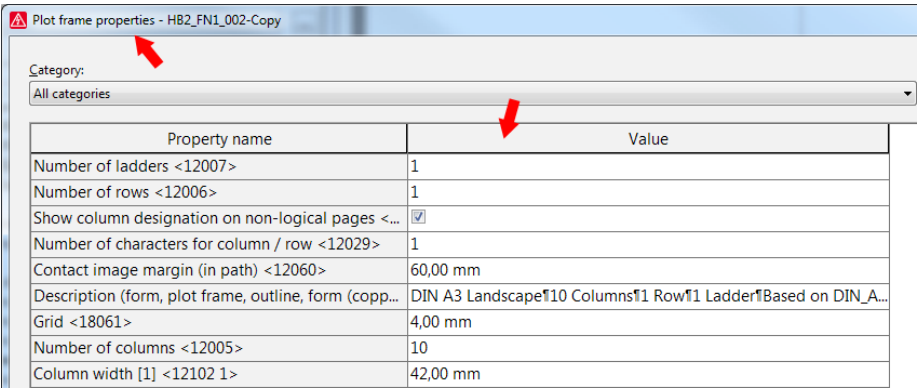


Fig. 10.44
Plot frame properties

Once all entries have been made or the plot frame has been graphically edited as desired, it can be closed (“saved”) by selecting, in the UTILITIES/MASTER DATA/PLOT FRAME menu, the CLOSE function.

EPLAN closes and permanently “saves” the plot frame in the system master data. If the plot frame is needed for the project, it will be automatically stored in the project from the system master data. This functional process is identical to working with forms; the same is true of the synchronization in case of subsequent modifications to the plot frame.

10.4.2 Edit existing plot frame

In addition to copying plot frames, existing plot frames can also be edited. To do this, select the UTILITIES/MASTER DATA/PLOT FRAME menu and then select the OPEN function. EPLAN opens the **Open plot frame** dialog.

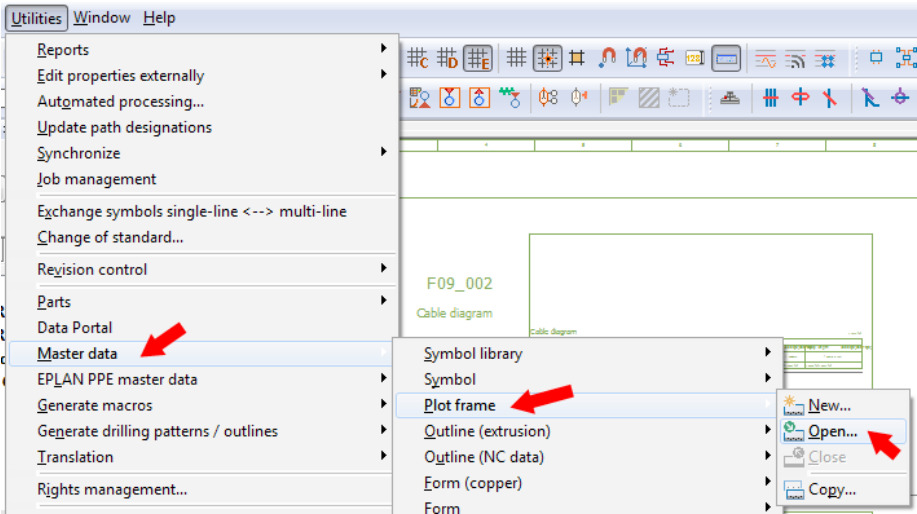


Fig. 10.45
Open plot frame dialog

Here you select the plot frame you wish to edit and open it via the **OPEN** button in the lower area of the dialog. EPLAN closes the dialog and opens the plot frame again as a temporary page in the page navigator of the current project.

It can now be edited using the familiar functions (graphics, text, etc.); the plot frame properties can also be edited or extended.

Close the plot frame using the **UTILITIES/MASTER DATA/PLOT FRAME/CLOSE** menu. If the plot frame has been used in the current project, you will be prompted whether you wish to synchronize with the project master data.

10.4.3 Create new plot frame

Instead of copying or editing existing plot frames, you can create a completely new plot frame via the **NEW** function in the **UTILITIES/MASTER DATA/PLOT FRAME** menu. EPLAN then opens the **Create plot frame** dialog.

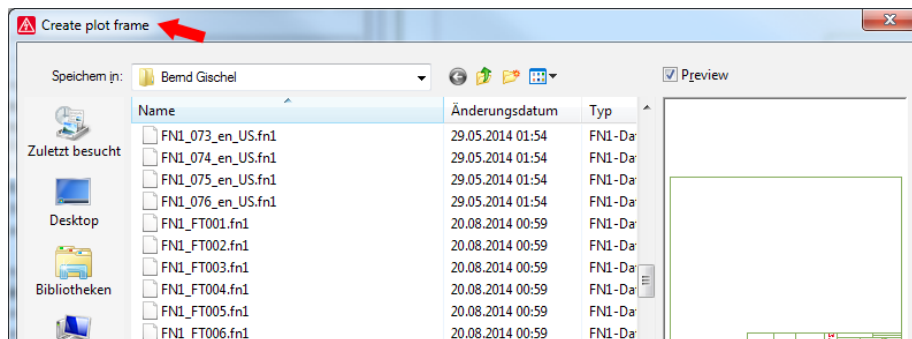


Fig. 10.46 Create plot frame dialog

You enter the new name of the plot frame in the **File name** field of the **Create plot frame** dialog.

Clicking the **SAVE** button creates the plot frame and EPLAN then opens the **plot frame properties** of the new plot frame and also displays an empty temporary page in the page navigator of the current project.

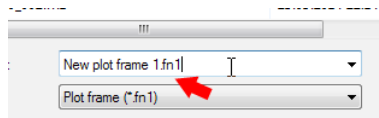


Fig. 10.47
New plot frame name
in the File name field

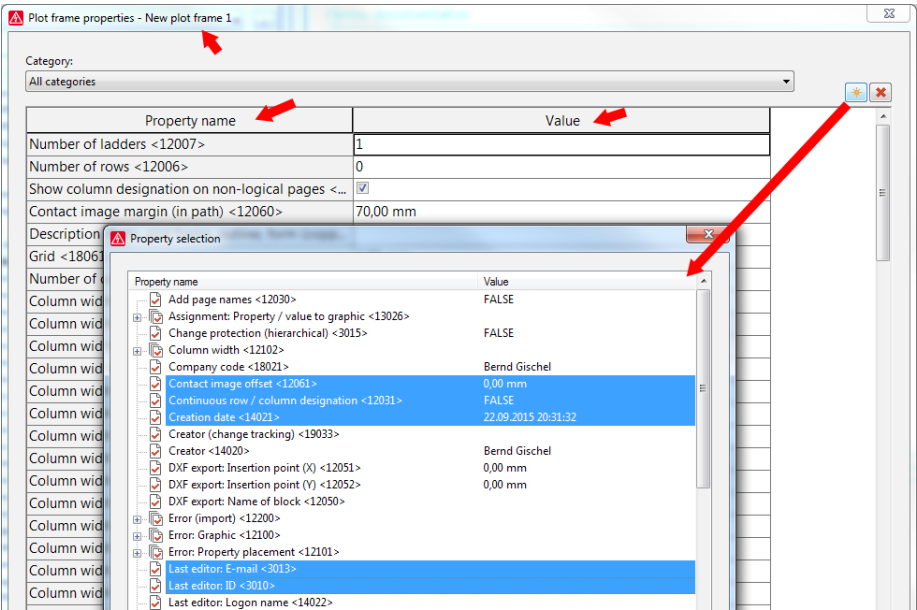


Fig. 10.48 Plot frame properties of the new plot frame

Now, you can edit the plot frame using the familiar functions.

NOTE: Instead of creating a new plot frame, you can also use a copied plot frame as a template for a new plot frame and then modify it accordingly.

This is done in the same way as editing an existing plot frame and will not be explained further here.

Generally speaking, importing data from older versions of EPLAN (EPLAN 5, EPLAN 21, PPE and FLUID) is a fast, time-saving way of continuing to work with these projects in EPLAN Electric P8. Conversion is relatively trouble-free, barring a few specific solutions. But it is still not the best solution.



TIP: It is a better to start new projects using EPLAN Electric P8 and not to base them on data from projects created with older versions.

It is also not absolutely necessary to convert all projects from old versions to EPLAN Electric P8. Since EPLAN will continue to support the earlier versions for a long time to come, there is no reason to panic. Older or currently running projects can still be edited or finished with the earlier versions.



NOTE: EPLAN supports the import of data from old versions of EPLAN 21, EPLAN PPE and EPLAN Cabinet up to EPLAN Electric P8 Version 2.3. From Version 2.4, this direct data import into the EPLAN Electric P8 interface will no longer be available.

This chapter only provides a very brief introduction to the import functions. More information can be found in the first edition of the reference handbook (ISBN 978-3-446-42266-9, only available in German), which has a detailed chapter on importing old data. Alternatively, you can use EPLAN Electric P8 online help, read the support documents on importing old EPLAN data to EPLAN Electric P8 found in the EPLAN support area, or use the many EPLAN Internet forums.

It is important to realize that there were numerous ways to create schematics in the old versions of EPLAN. It is therefore unavoidable that deviations will occur here and there in a converted (imported) project.



NOTE: If projects are imported into EPLAN Electric P8 from an old version, then the old project remains intact. After importing, you have two projects, the original project (that of the old version) and the same project (“new”) converted to the EPLAN Electric P8 data model. None of the old data is deleted or changed, so the original project can still be edited in the previous version without problems.

If a project is edited further in previous versions, then it is no longer the same as the project that was previously converted to an EPLAN Electric P8 project. Therefore it must be re-imported into EPLAN Electric P8.

■ 11.1 Import options

EPLAN Electric P8 offers an import assistant for the various old versions of EPLAN 5, as well as for the different data from older versions. This can encompass entire projects or individual data sets, such as a form, a macro, parts data or cable data.

You will find this import assistant in the **UTILITIES/DATA TRANSFER** menu in the **EPLAN 5/FLUIDPLAN** data transfer menu items (for EPLAN 5 and Fluid projects, as well as for EPLAN 5/Fluid master data).

In general, it is possible to import both entire projects and individual data using EPLAN Electric P8. Additional functions and calls were integrated into the individual data import menu items (EPLAN 5/fluidPLAN) to enable you, for example, to import into EPLAN Electric P8 an individual plot frame or parts data exported from the older system.

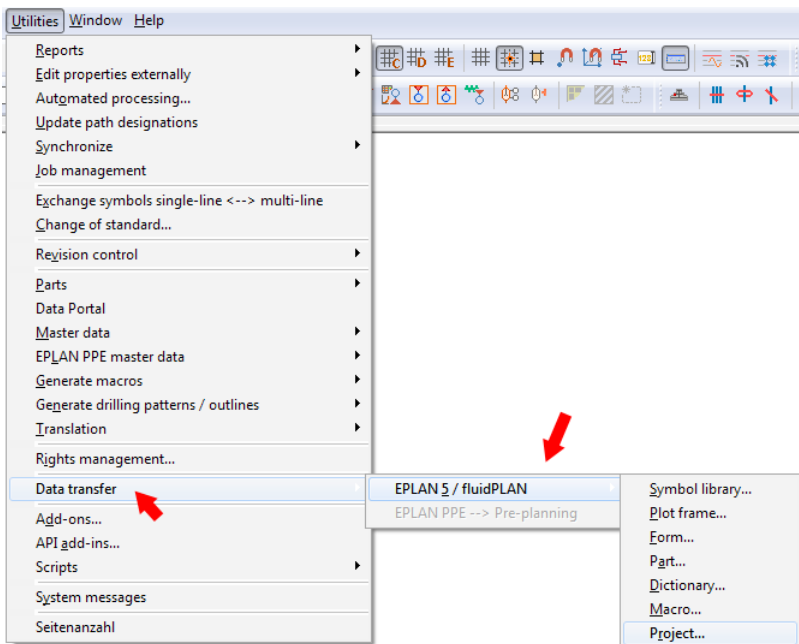


Fig. 11.1 Data transfer menu



NOTE: The sequence of menu items in the menu is important during data importation. This sequence was not arbitrarily chosen; it represents the standard sequence of a data import.

An EPLAN 5 data import, for example, should begin with the symbol library. Then the plot frame should be imported followed by the forms, the parts data and the foreign language file (the dictionary in EPLAN Electric P8), and any macros. At the end, the actual EPLAN 5 project can be imported.

In addition to these individual steps, EPLAN Electric P8 offers a “one-click wizard” for importing data. To import old data, you start with the UTILITIES/DATA IMPORT/EPLAN 5 – FLUIDPLAN/PROJECT menu item, enter some information in the subsequent dialogs, and then let EPLAN Electric P8 automatically take care of the rest of the import procedure.

In addition to the standard functions delivered with EPLAN Electric P8, there are a series of further function extensions that are covered by additional modules, called add-ons. These add-ons make it easier to work with projects and their data, or help in other ways to accelerate and optimize engineering. This chapter will explain some of these add-ons. In addition to those that are discussed here, there are also a series of further add-ons that can be requested directly from EPLAN.

■ 12.1 EPLAN Data Portal

The previous chapters mentioned the necessity of correct function templates (the individual function definitions of devices), device selection and finished macros. Function templates, among other things, are a major focus when working with EPLAN Electric P8. Users could, of course, fill their own parts with this additional information themselves, or they can use the EPLAN Data Portal solution.

12.1.1 What are the advantages of the EPLAN Data Portal?

The EPLAN Data Portal provides online access to parts data that can be directly used during current and subsequent project planning. This parts data is usually provided by the manufacturer, is subjected to an incoming inspection by EPLAN (whereby EPLAN as a software manufacturer cannot, of course, check the correctness of the data provided by the manufacturer), and is then provided online in the EPLAN Data Portal. This allows users to directly use parts (devices, macros, etc.) from the EPLAN Data Portal, without having to add them to, or create them in their own parts management.